

KMOU Division of Marine System Engineering

Introduction to Internal Combustion Engine



국립 한국해양대학교
NATIONAL
KOREA MARITIME & OCEAN UNIVERSITY

Made by Minho Park (Ph.D.)

<https://orcid.org/0000-0002-6900-4566>

Lesson content for each week

Ch0_Introduction to the Class

Ch6_Piston Fittings and Crankshaft

Ch1_Overview of the Internal Combustion Engine

Ch7_Lube Oil System

Ch2_Thermodynamics and Compressed Air System

Ch8_Cooling Water System

Ch3_Marine Fuels and Internal Fuel Oil System

Ch9_Introduction to the Main Engine

Ch4_Intake and Exhaust System

Ch10_Alternative Fuels and Alternative Fuel Systems

Ch5_Turbocharger

Ch11_Environmental Regulations and Exhaust Gas Aftertreatment Systems

Midterm

Ch12_Tools, Apprentice Engineer Duties, and Career Direction

Final Exam

Initial starting of the G/E

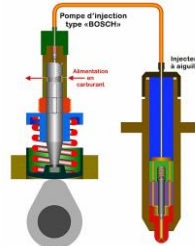
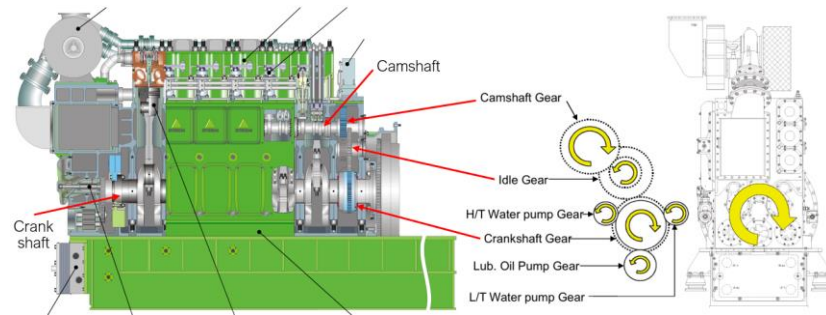
The crankshaft connected to the flywheel is rotated by the G/E's air starter.



Accordingly, the piston connected to the crankshaft performs a reciprocating motion.



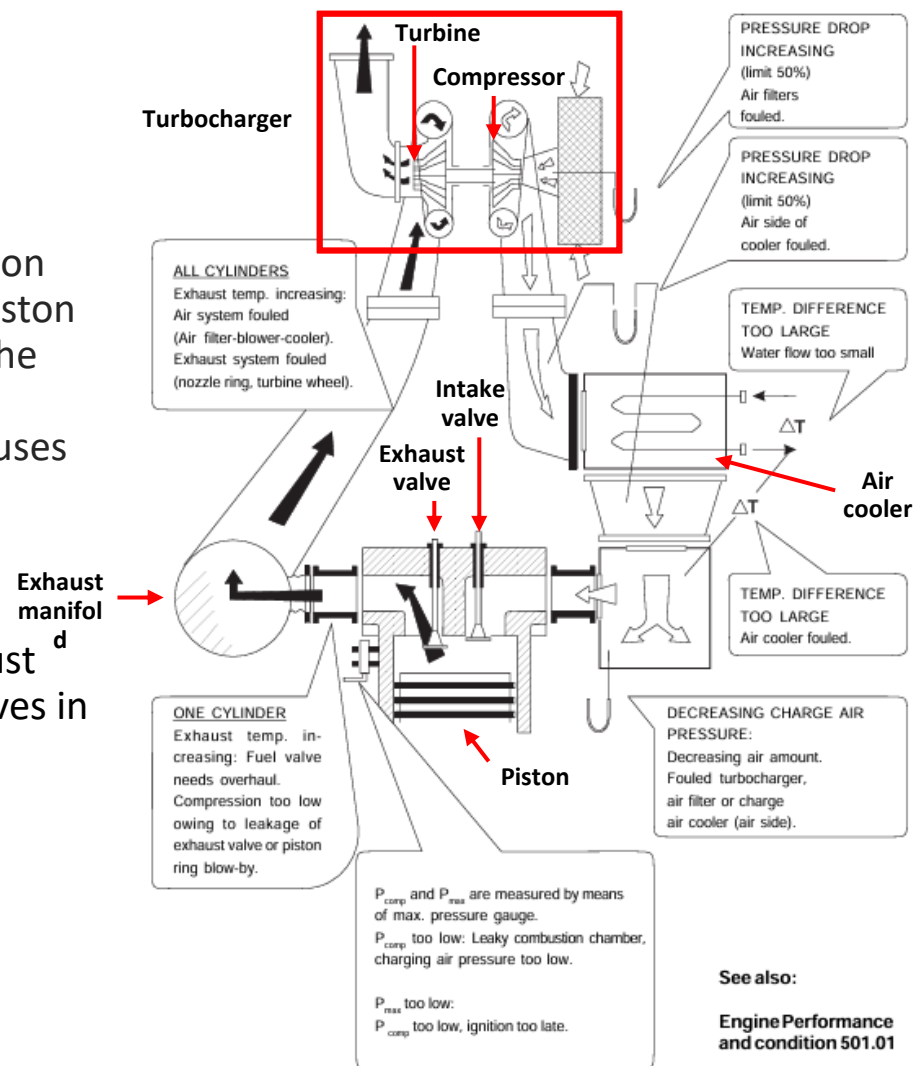
Simultaneously, as the crankshaft rotates, the camshaft, which is connected to the gears of the crankshaft, rotates. As the camshaft rotates, the fuel injection pump compresses the fuel by the cam, and the atomized fuel is injected into the cylinder through the high oil pressure pipe and the fuel injection valve, when the fuel rack is positioned in a position where fuel injection is possible.



Intake & exhaust system

The fuel injected into the cylinder and the compression heat generated by the reciprocating motion of the piston cause the injected fuel to mix instantaneously with the high-temperature, high-pressure air, resulting in spontaneous ignition, and the resulting explosion causes the piston to reciprocate and produces exhaust gas.

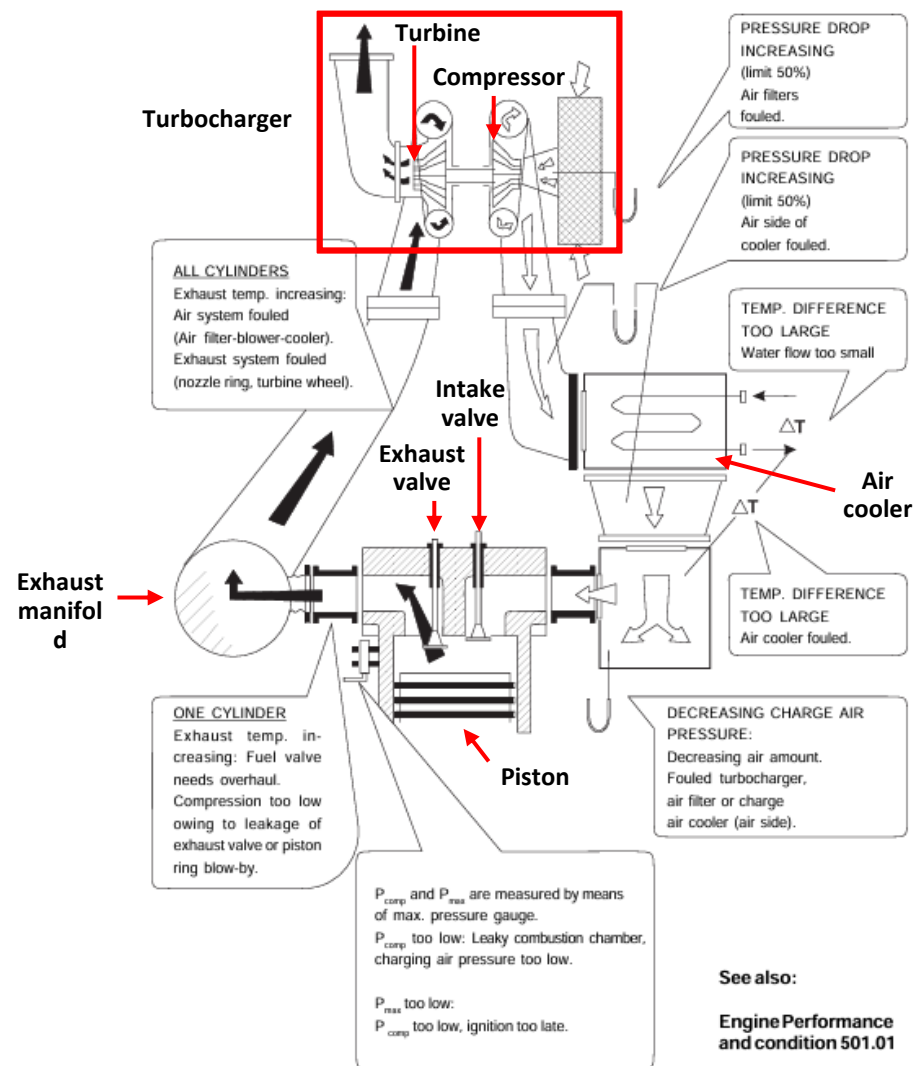
The generated exhaust gas passes through the exhaust manifold to the turbocharger as the two exhaust valves in the cylinder head open.



Intake & exhaust system

The exhaust gas rotates the turbine of the turbocharger, which in turn rotates the compressor connected to the turbine by a shaft, compressing the outside air and supplying the compressed air into the cylinder through the air cooler as the intake valve opens, contributing to the explosion occurring inside the cylinder.

During initial startup of a G/E, the amount of fuel injected under no-load conditions is small, and the intake air pressure is low. This results in insufficient heat energy from the exhaust gas generated by the explosion. Consequently, the rotational speeds of the turbocharger turbine and compressor are low, ultimately reducing the pressure of the intake air supplied to the cylinder.



Intake & exhaust system

The low intake air pressure during the initial start-up of G/E causes the following problems:

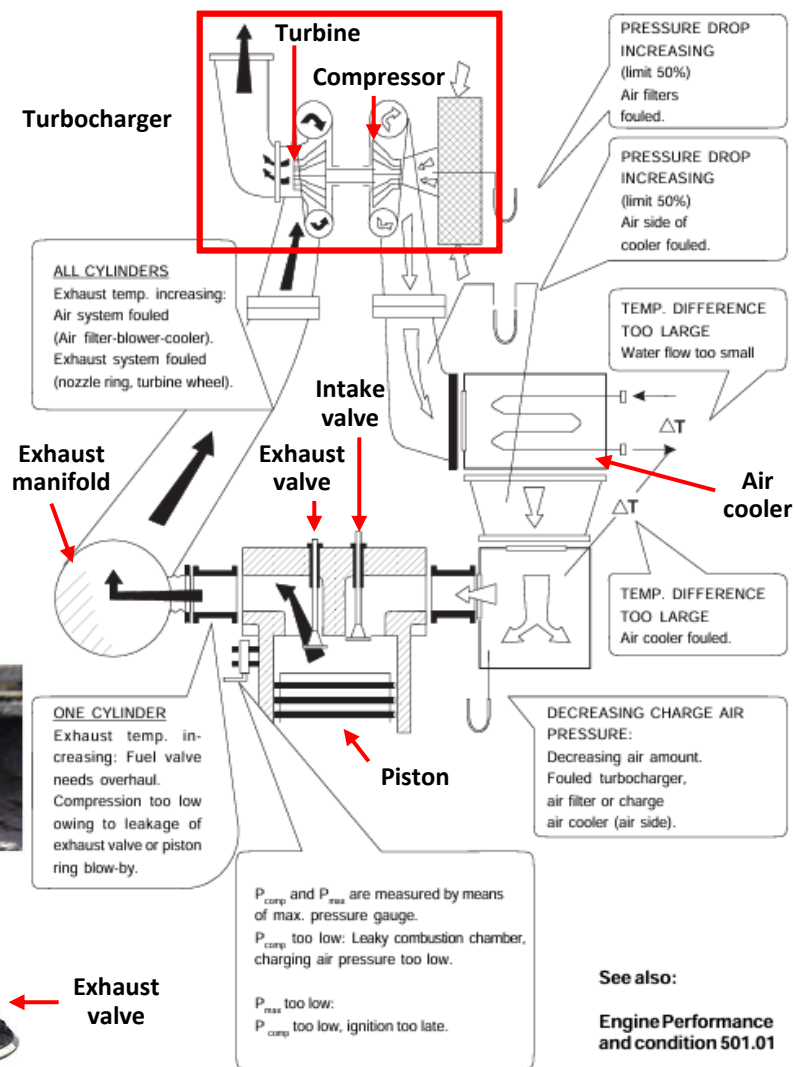
- 1) Incomplete combustion due to low-load combustion
- 2) Black smoke caused by incomplete combustion



- 3) Fouling of the turbine side of the turbocharger (T/C) caused by black smoke



- 4) Fouling in the exhaust system of the G/E caused by black smoke



Intake & exhaust system

The picture on the right is for reference only.

In the previous slide, the intake side and exhaust side are shown as being located at the front and rear of the cylinder head, but this is for understanding purposes only.

In reality, the intake and exhaust systems are located at the rear of the cylinder head, as shown in the picture on the right.

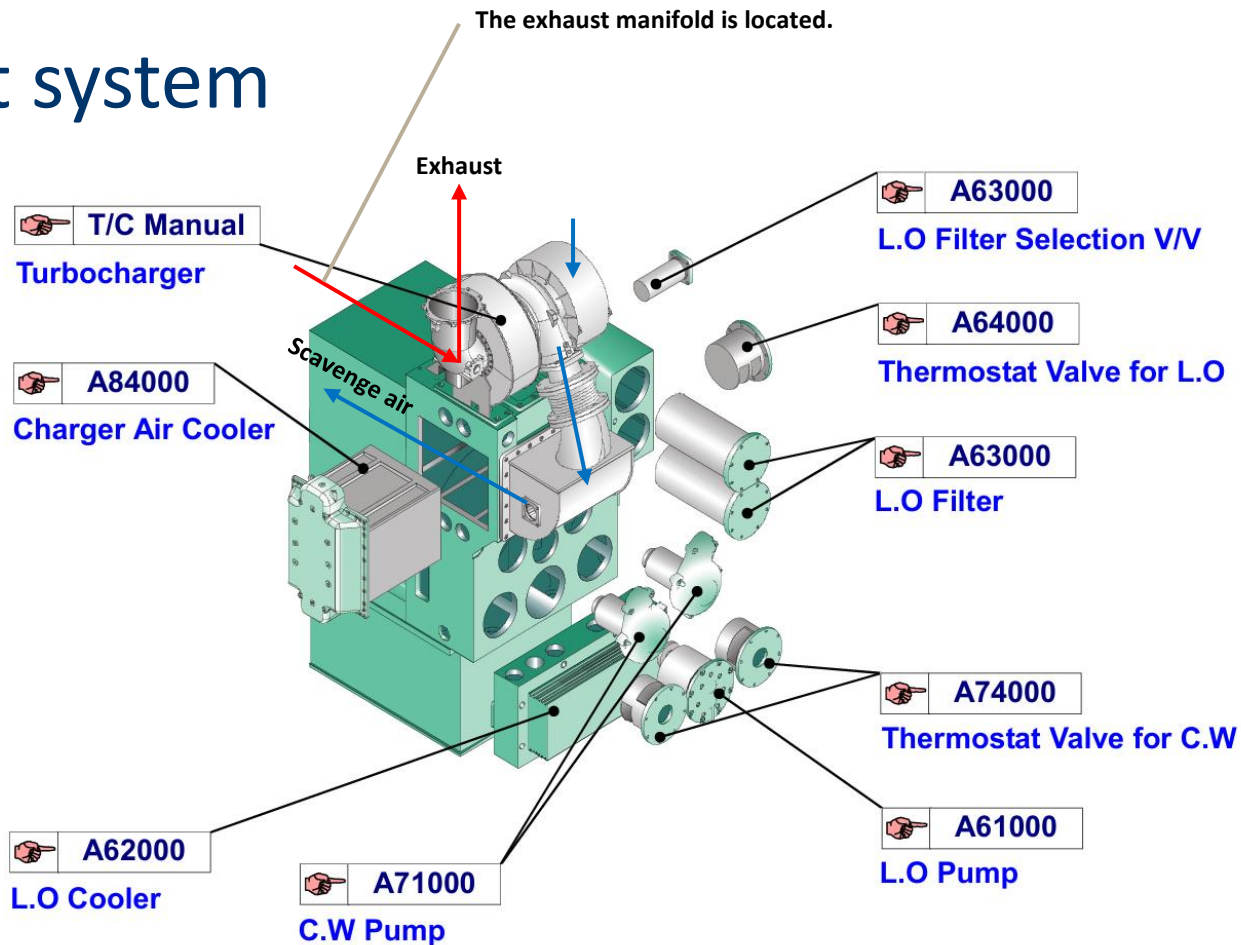


Fig. 2 Equipments for feed module (피드 모듈 장치)

Intake & exhaust system

Recently manufactured G/Es use a jet system to improve the problem of low intake air pressure during initial startup.

The jet system supplies compressed air to the turbocharger's turbine only for a certain period of time during initial startup to assist turbine rotation. Accordingly, it contributes to increasing the pressure of the intake air.

Plate Page: 1 (2)	ON-OFF Valve for Jet System	51320-01H
----------------------	-----------------------------	-----------

L23/30H

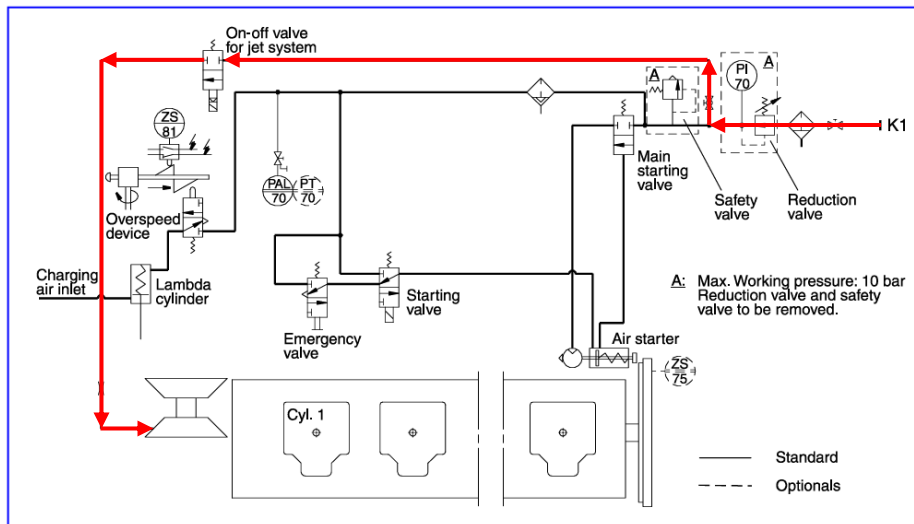
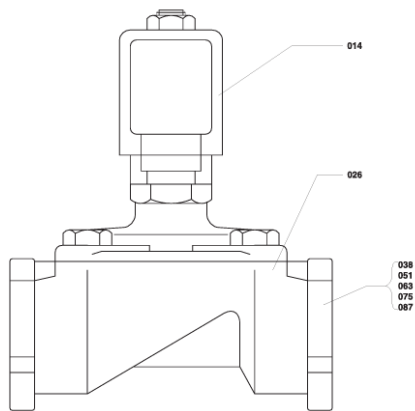
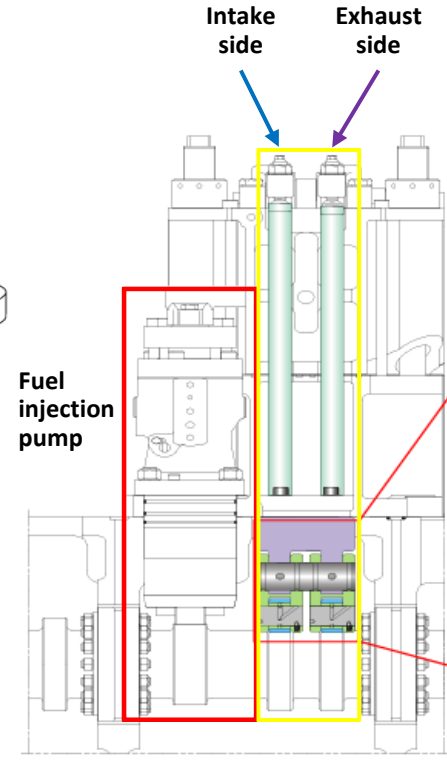
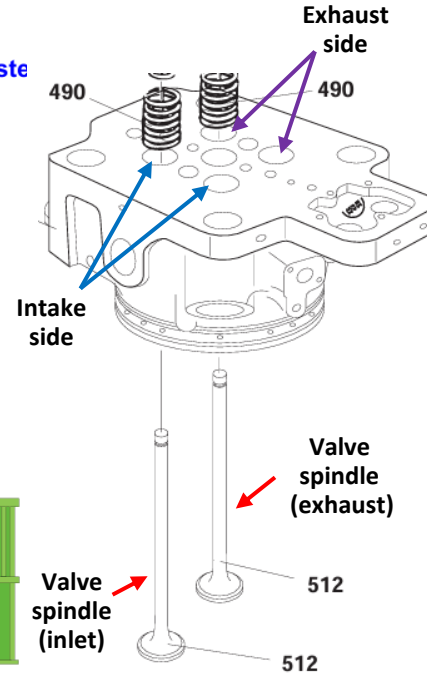
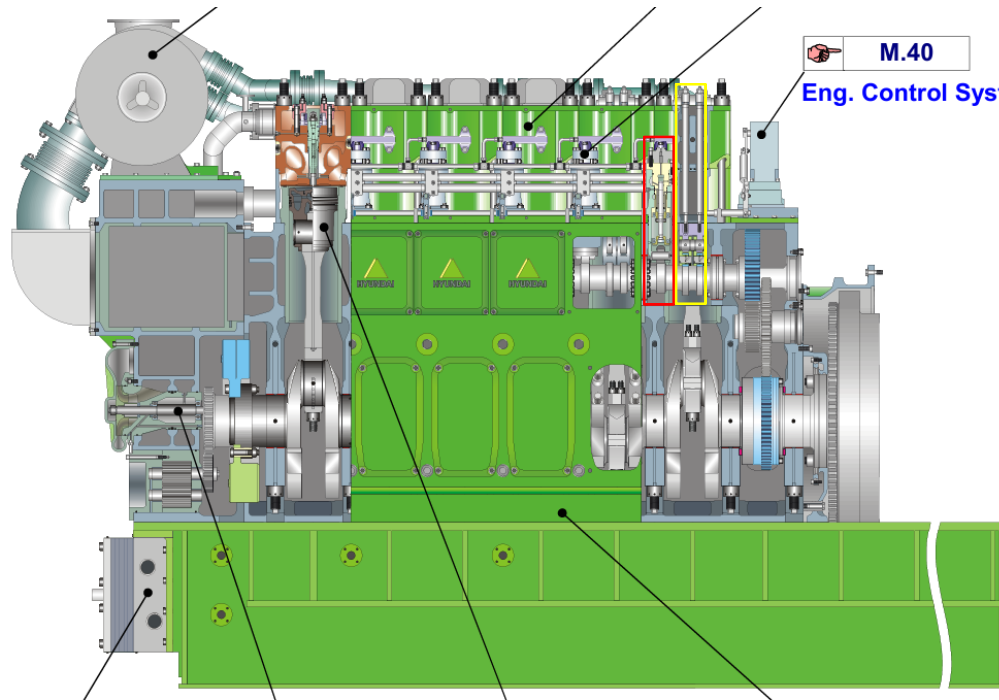


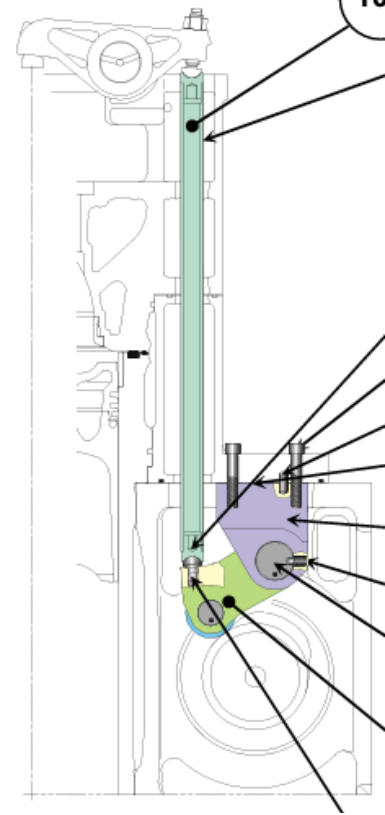
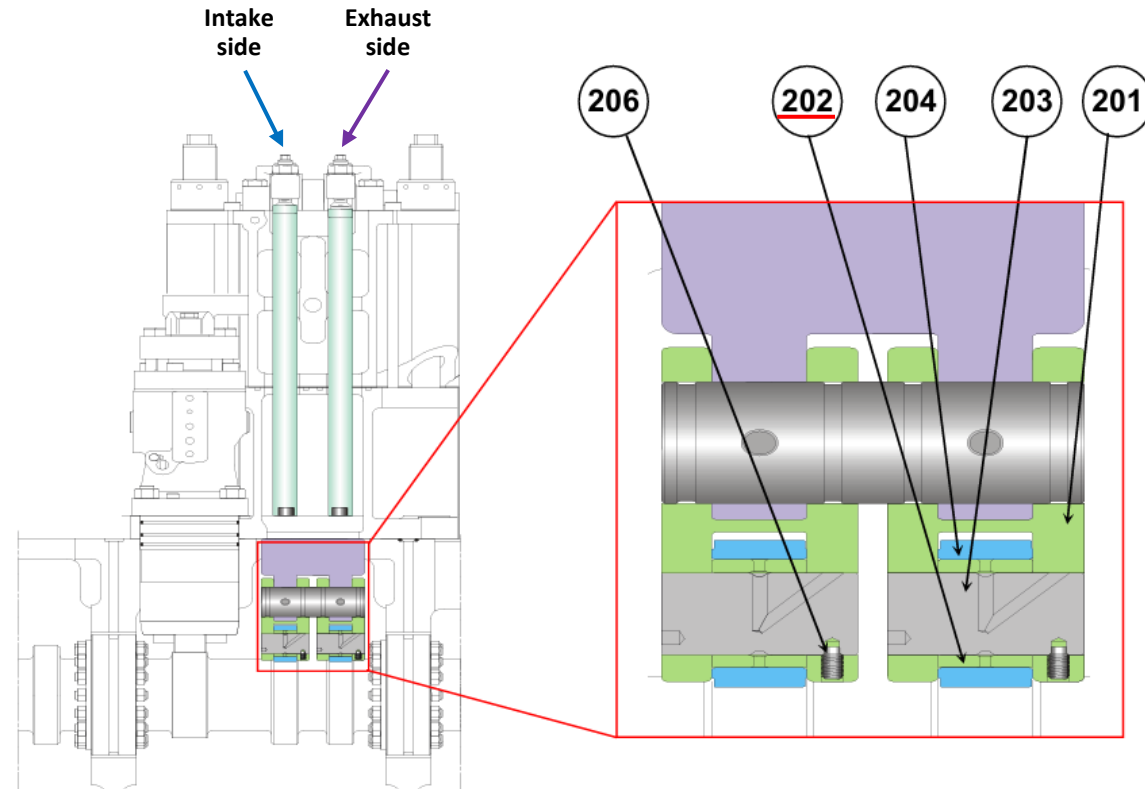
Fig 1 Diagram for compressed air system.

Intake & exhaust system

The two intake valves and two exhaust valves mounted on the cylinder head of the G/E are both operated by two cams on the camshaft.



Intake & exhaust system



Part List

Item No.	Description	Q'ty / Cyl.
201	Swing arm body	2
<u>202</u>	<u>Roller</u>	2
203	Roller pin	2
204	Roller bush	2
206	Set screw	2

Intake & exhaust system

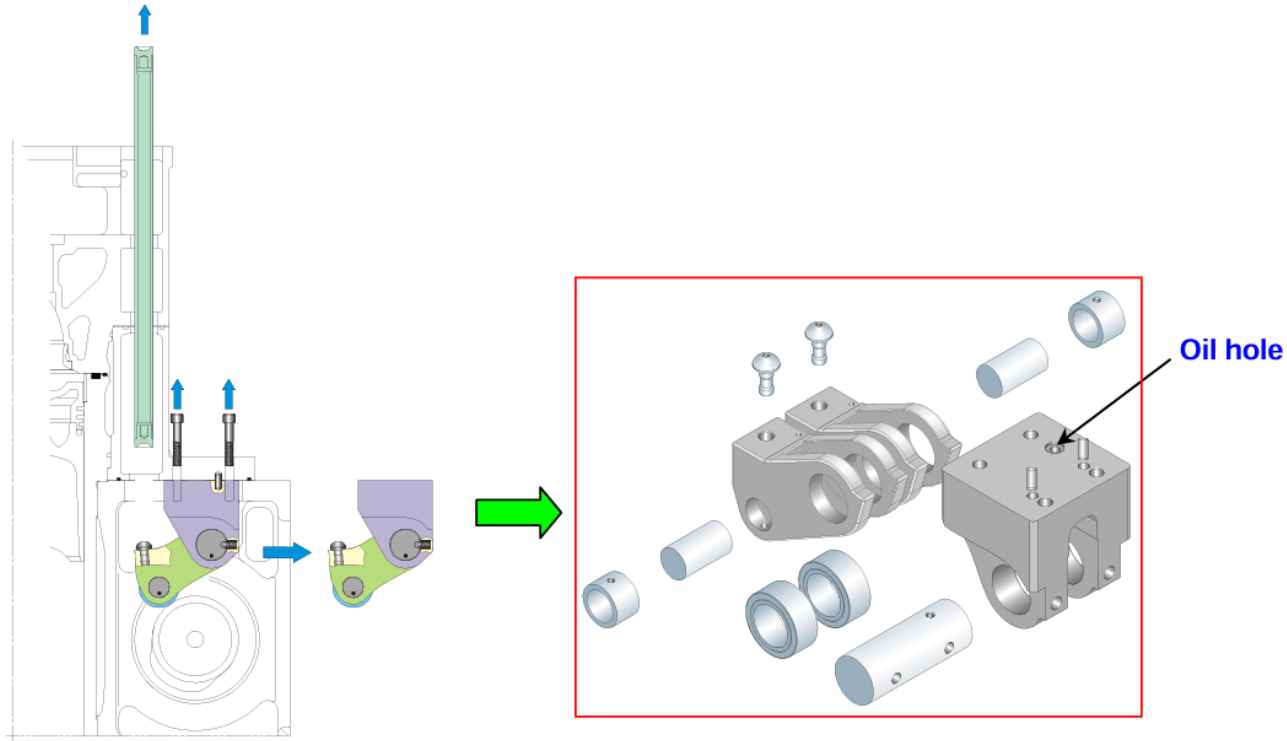
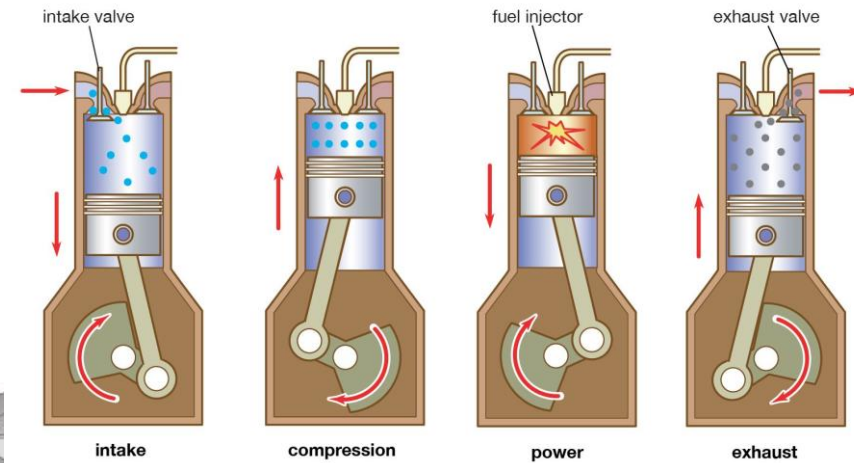


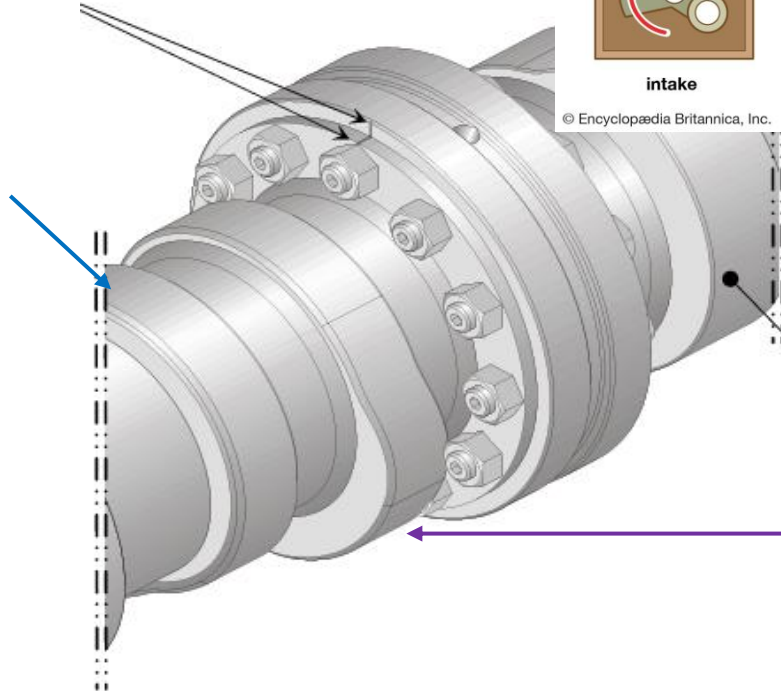
Fig. 1 Dismantling of a swing arm assembly (스윙 암 어셈블리의 분해)

Intake & exhaust system



© Encyclopædia Britannica, Inc.

Cam for intake side



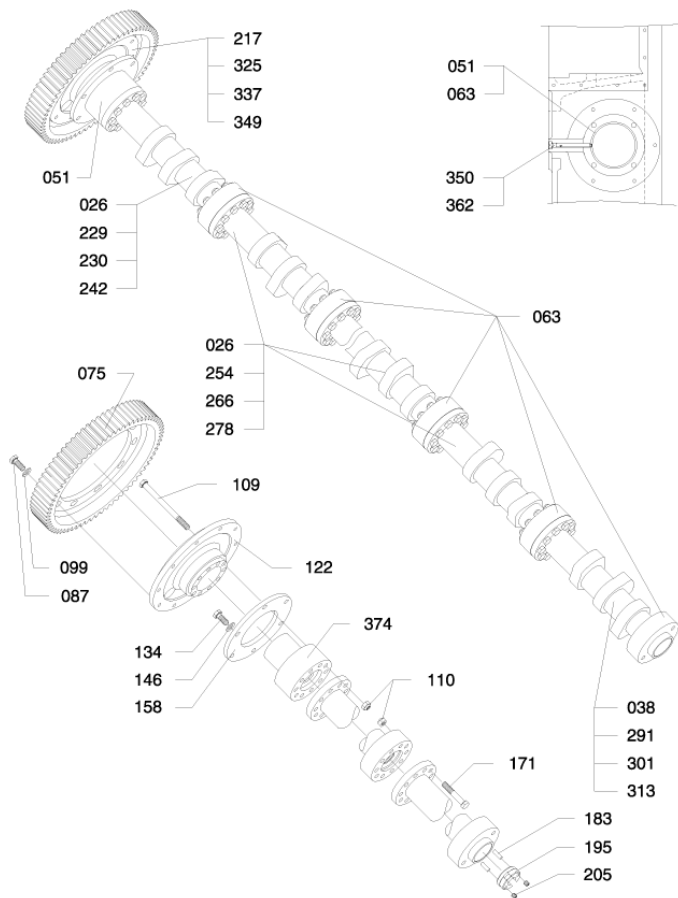
Because the convex portions of the cams on the intake and exhaust sides are different, the intake and exhaust valves open at different times.

Cam for exhaust side



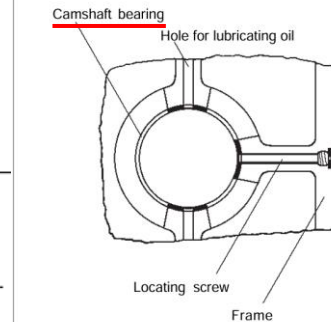
한국해양대학교
NATIONAL KOREA MARITIME & OCEAN UNIVERSITY

Camshaft and Camshaft Bearing

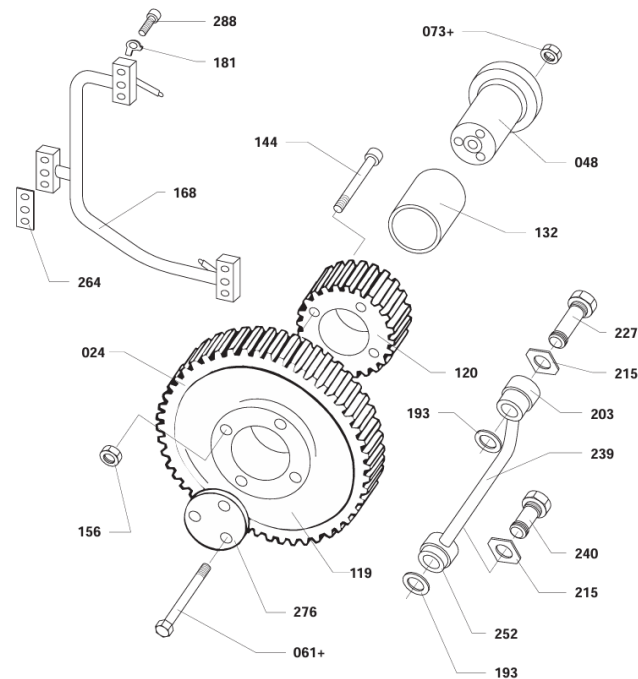
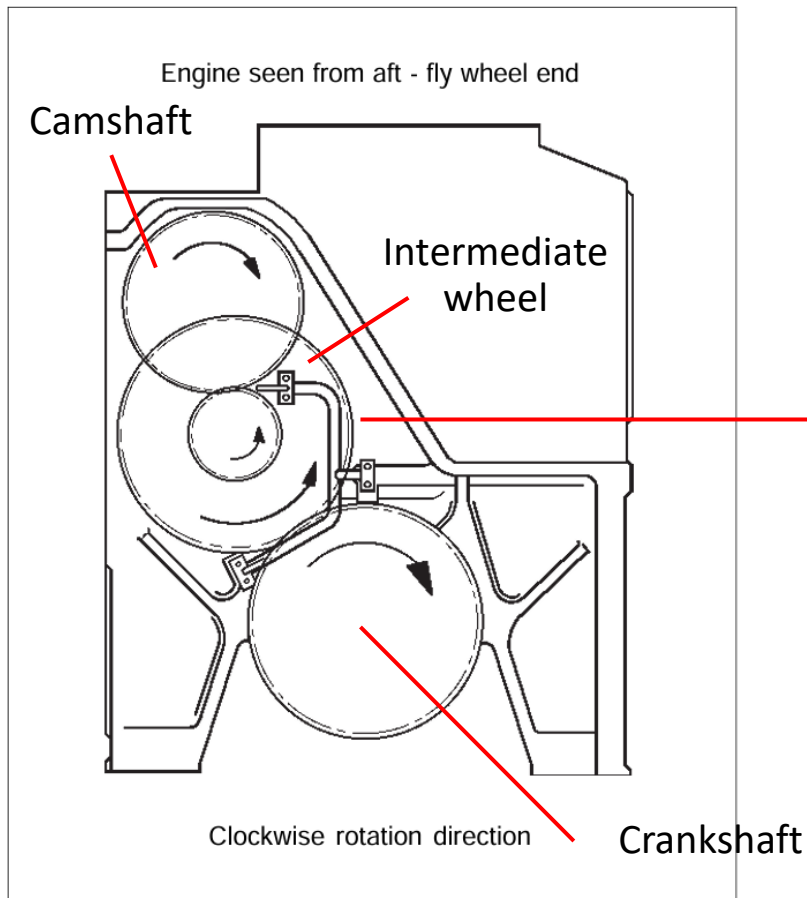


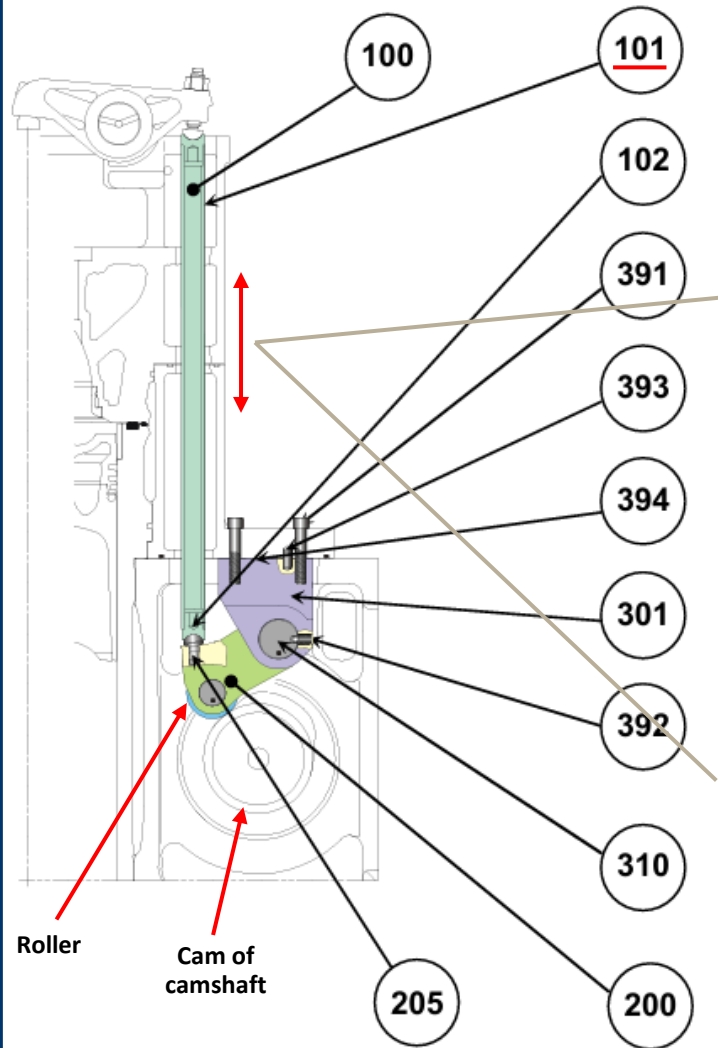
Item no	Qty.	Designation	Benaevnelse
026	4/E	Camshaft, aft/intermediate, 5 cyl. engine	Styreksel agter/mellem, 5 cyl. motor
038	1/E	Camshaft "fore" 5 cyl. engine	Styreksel "for" 5 cyl. motor
051	1/E	Camshaft bearing	Styrekselleje
063	5/E 6/E 7/E 8/E	Camshaft bearing	Styrekselleje 5 cyl. motor 6 cyl. motor 7 cyl. motor 8 cyl. motor
075	1/E	Gear wheel	Tandhjul
087	10/E	Screw	Skrue
099	10/E	Spring washer	Fjederskive
109	10/E	Screw	Skrue
110	50/E 40/E 52/E 66/E	Self locking nut 5 cyl. engine 6 cyl. engine 7 cyl. engine 8 cyl. engine	Selvlåsende møtrik 5 cyl. motor 6 cyl. motor 7 cyl. motor 8 cyl. motor
122	1/E	Hub	Nav
134	6/E	Screw	Skrue
146	6/E	Disc spring	Fjederskive
158	1/E	Guide ring	Sikringsring
171	40/E 30/E 42/E 56/E	Screw 5 cyl. engine 6 cyl. engine 7 cyl. engine 8 cyl. engine	Skrue 5 cyl. motor 6 cyl. motor 7 cyl. motor 8 cyl. motor
183	2/E	Spring pin	Fjedertap
195	1/E	Coupling	Kobling
205	2/E	Screw	Skrue
217	1/E	Camshaft, complete for 5 cyl. engine, incl. item 026, 038, 109, 110, 122, 134, 146, 158, 171, 183, 195, 205, 374	Styreksel komplet, 5 cyl. motor, inkl. item 026, 038, 109, 110, 122, 134, 146, 158, 171, 183, 195, 205, 374
229	1/E	Camshaft "aft" 6 cyl. engine	Styreksel, agter 6 cyl. motor

Item no	Qty.	Designation	
230	1/E	Camshaft "aft" 7 cyl. engine	
242	1/E	Camshaft "aft" 8 cyl. engine	
254	4/E	Camshaft, intermediate, 6 cyl. engine	
266	5/E	Camshaft, intermediate, 7 cyl. engine	
278	6/E	Camshaft, intermediate, 8 cyl. engine	Styreksel mellem 8 cyl. motor
291	1/E	Camshaft "fore" 6 cyl. engine	Styreksel "for" 6 cyl. motor
301	1/E	Camshaft "fore" 7 cyl. engine	Styreksel "for" 7 cyl. motor
313	1/E	Camshaft "fore" 8 cyl. engine	Styreksel "for" 8 cyl. motor
325	1/E	Camshaft complete, 6 cyl. eng. incl. item 109, 110, 122, 134, 146, 158, 171, 183, 195, 205, 229, 254, 291, 374	Styreksel komplet, 6 cyl. motor inkl. item 109, 110, 122, 134, 146, 158, 171, 183, 195, 205, 229, 254, 291, 374
337	1/E	Camshaft complete, 7 cyl. eng. incl. item 109, 110, 122, 134, 146, 158, 171, 183, 195, 205, 230, 266, 301, 374	Styreksel komplet, 7 cyl. motor inkl. item 109, 110, 122, 134, 146, 158, 171, 183, 195, 205, 230, 266, 301, 374
349	1/E	Camshaft complete, 8 cyl. eng. incl. item 109, 110, 122, 134, 146, 158, 171, 183, 195, 205, 242, 278, 313, 374	Styreksel komplet, 8 cyl. motor inkl. item 109, 110, 122, 134, 146, 158, 171, 183, 195, 205, 242, 278, 313, 374
350	6/E 7/E 8/E 9/E	Guide screw 5 cyl. engine 6 cyl. engine 7 cyl. engine 8 cyl. engine	Styreskrue 5 cyl. motor 6 cyl. motor 7 cyl. motor 8 cyl. motor
362	6/E 7/E 8/E 9/E	Packing ring 5 cyl. engine 6 cyl. engine 7 cyl. engine 8 cyl. engine	Pakningsring 5 cyl. motor 6 cyl. motor 7 cyl. motor 8 cyl. motor
374	1/E	Shaft	Aksel

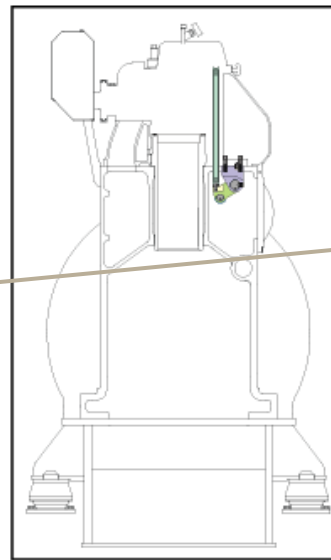


Camshaft





When the convex part of the camshaft ends and the flat part contacts the roller, the roller descends and the push rod descends accordingly.



The convex part of the camshaft causes the roller to rise, which in turn causes the push rod to rise as well.

Part List

Item No.	Description	Q'ty / Cyl.
100	Push rod assembly	2
<u>101</u>	<u>Push rod</u>	2
102	Ball seat(push rod)	4
200	Swing arm assembly	2
205	Swing arm piece bolt	2
301	Swing arm support	1
310	Swing arm pin	1
391	Socket head bolt	4
392	Set screw	2
393	Parallel pin	2
394	O-ring	1

Intake & exhaust system (MAN B&W)

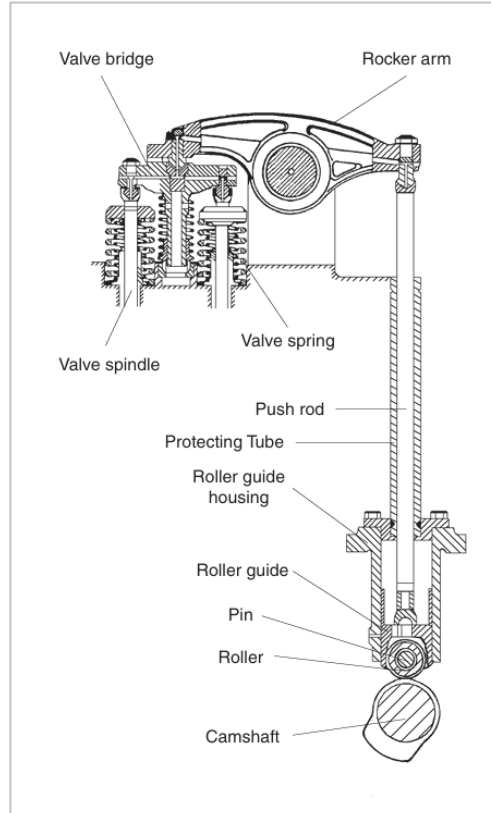
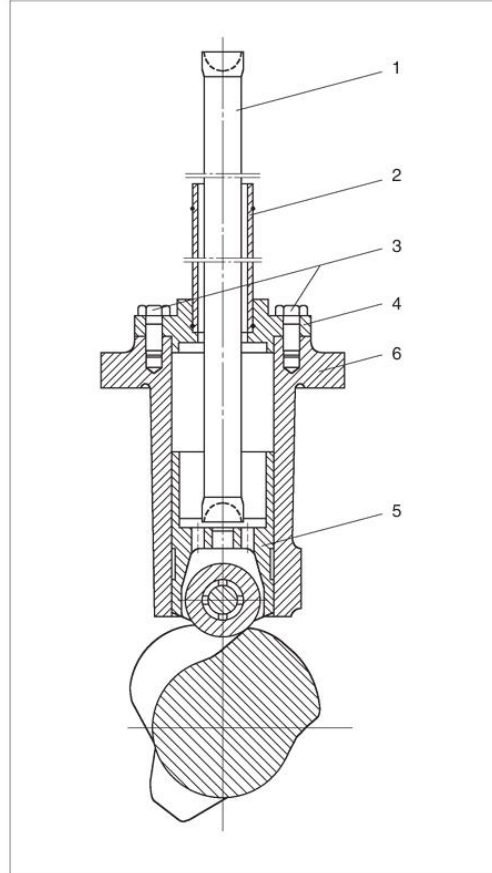
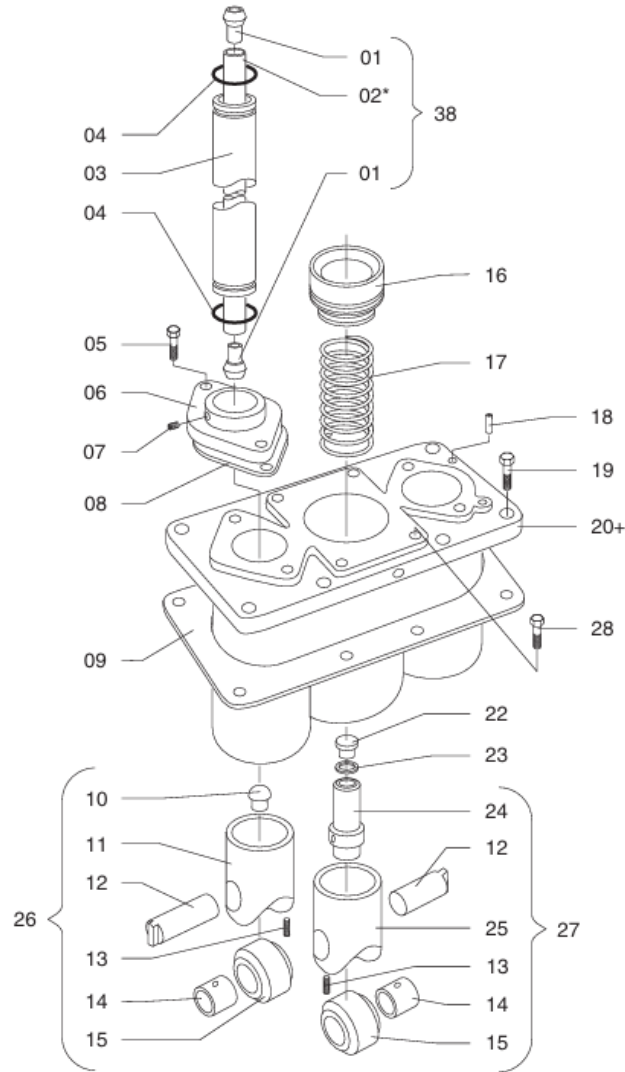


Fig. 1. Valve Operating Gear.

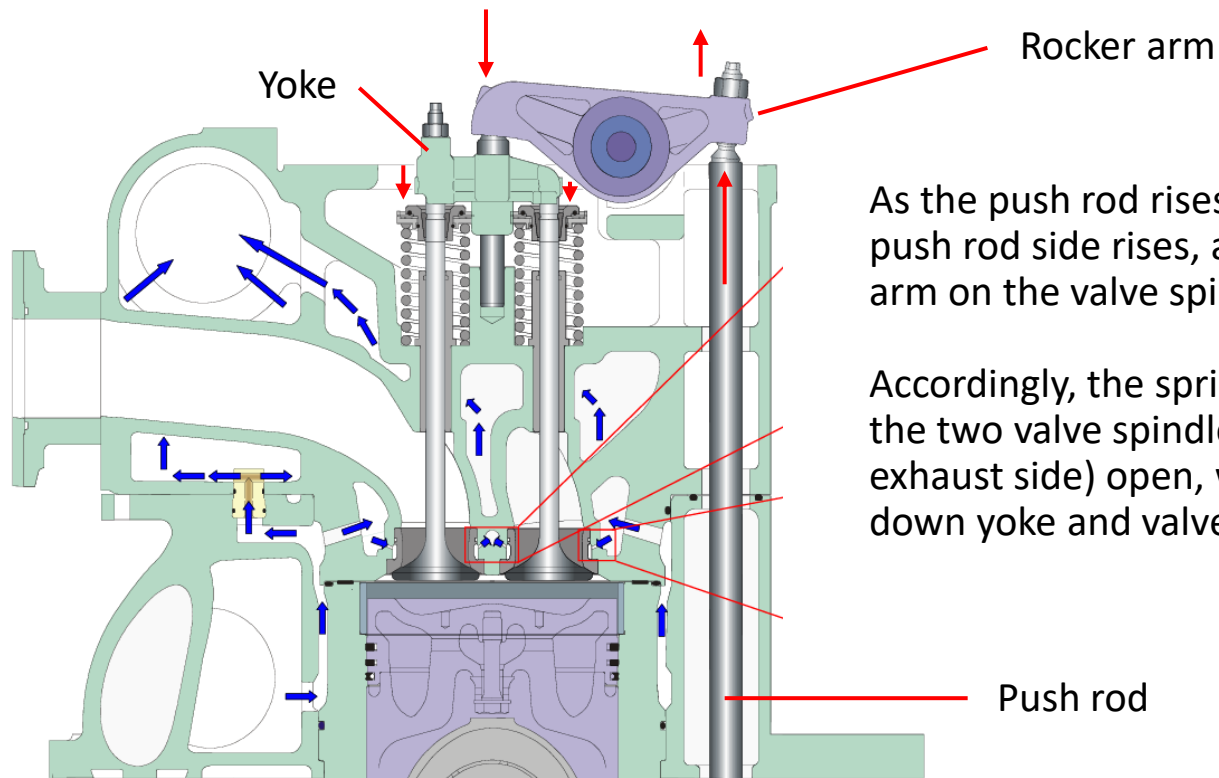




Item no	Qty.	Designation	Benævnelse
01	4/C	Thrust piece	Trykstykke
02*	2/C	Pipe	Rør
03	2/C	Protecting tube	Skærmrør
04	4/C	Sealing ring	Tætningsring
05	4/C	Screw	Skrue
06	2/C	Cover	Dæksel
07	2/C	Screw	Skrue
08	2/C	Gasket	Pakning
09	1/C	Gasket	Pakning
10	2/C	Ball pin	Kugletap
11	2/C	Roller guide	Rullestyr
12	3/C	Pin	Tap
13	3/C	Stop screw	Stopskrue
14	3/C	Bush	Foring
15	3/C	Roller	Rulle
16	1/C	Washer for spring	Skive for fjeder
17	1/C	Spring	Fjeder
18	2/C	Guide pin	Styrestift
19	8/C	Screw	Skrue
20+	1/C	Housing for roller guides	Hus for rullestyr
22	1/C	Thrust piece	Tryktap
23	1/C	1 set shims (0.1, 0.3, 0.5 and 1.0 mm)	1 sæt shims (0,1, 0,3, 0,5 og 1,0 mm)
24	1/C	Thrust piece	Tryktap
25	1/C	Roller guide	Rullestyr
26	2/C	Roller guide for valve gear, complete incl. item 10, 11, 12, 13, 14, 15	Rullestyr for ventilbevægelse, komplet inkl. item 10, 11, 12, 13, 14, 15

Item no	Qty.	Designation	Benævnelse
27	1/C	Roller guide for fuel injection pump, complete incl. item 12, 13, 14, 15, 24, 25	Rullestyr for brændselspumpe, komplet inkl. item 12, 13, 14, 15, 24, 25
28	4/C	Screw	Skrue
38	2/C	Push rod, complete incl. item 01, 02	Stødstang, komplet inkl. item 01, 02
		+ Item no 20 require an individual matching before mounting, contact MAN Diesel & Turbo, Høleby	+ Item nr. 20 kræver en individuel tilpasning før montering, kontakt MAN Diesel & Turbo, Høleby

Intake & exhaust system



As the push rod rises, the rocker arm on the push rod side rises, and conversely, the rocker arm on the valve spindle side goes down.

Accordingly, the spring force is overcome and the two valve spindles (either intake or exhaust side) open, when rocker arm pushes down yoke and valve spindle.

Fig.1 The cooling water flow in cylinder head (실린더 헤드내의 냉각수의 흐름)

Intake & exhaust system

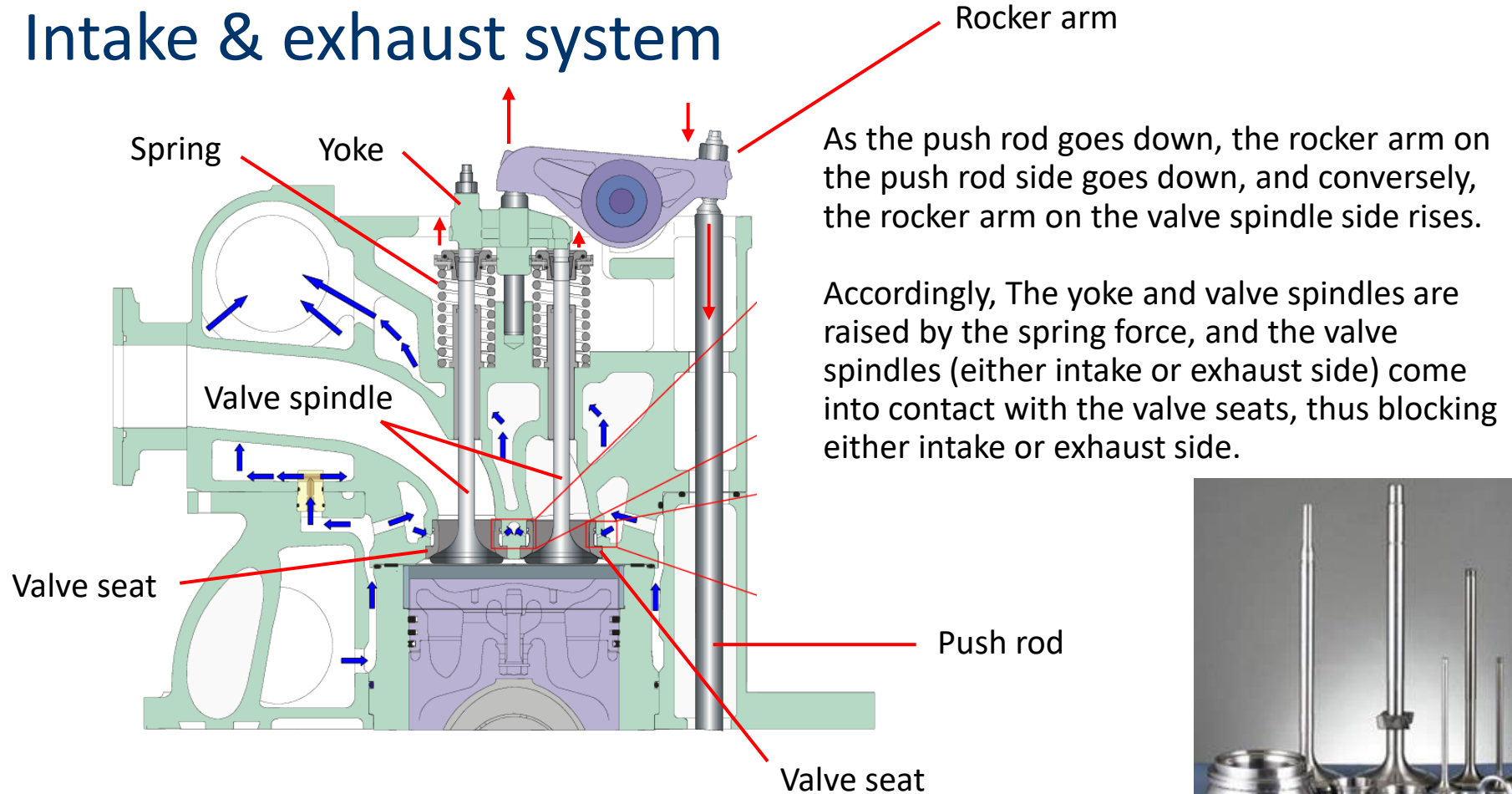
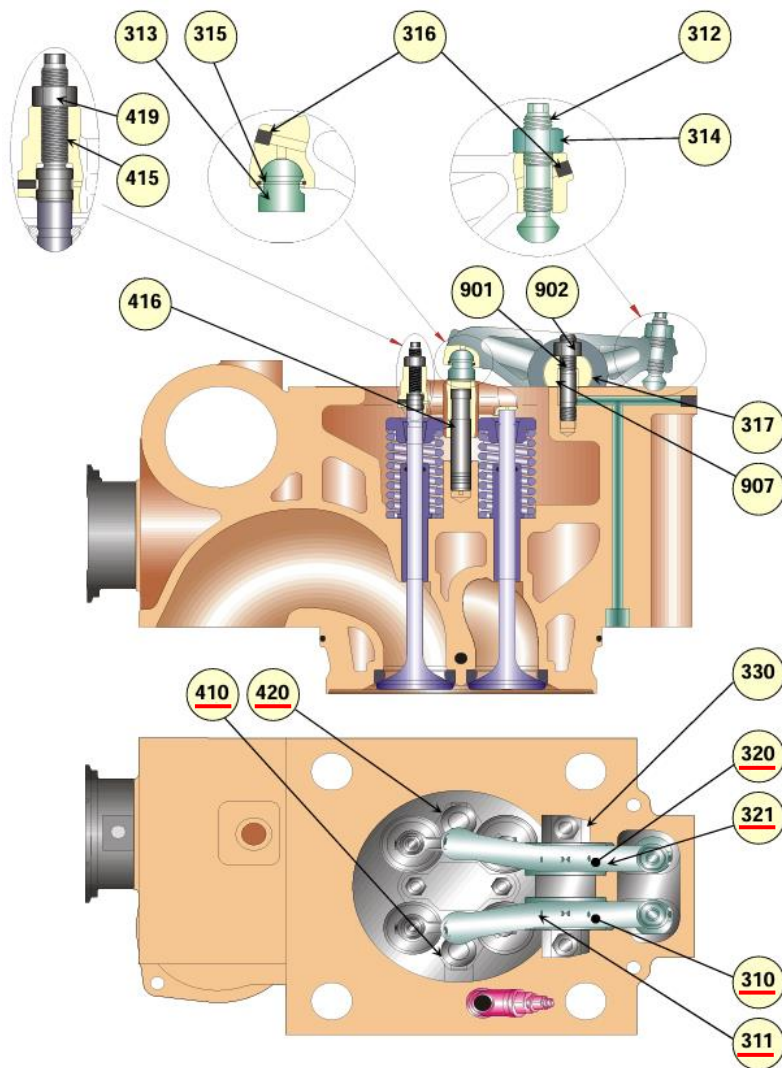


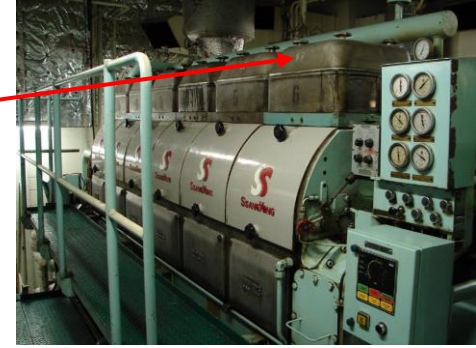
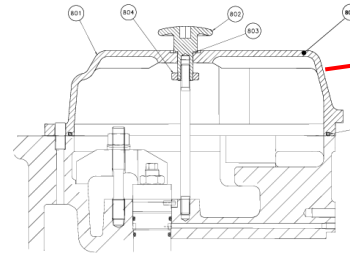
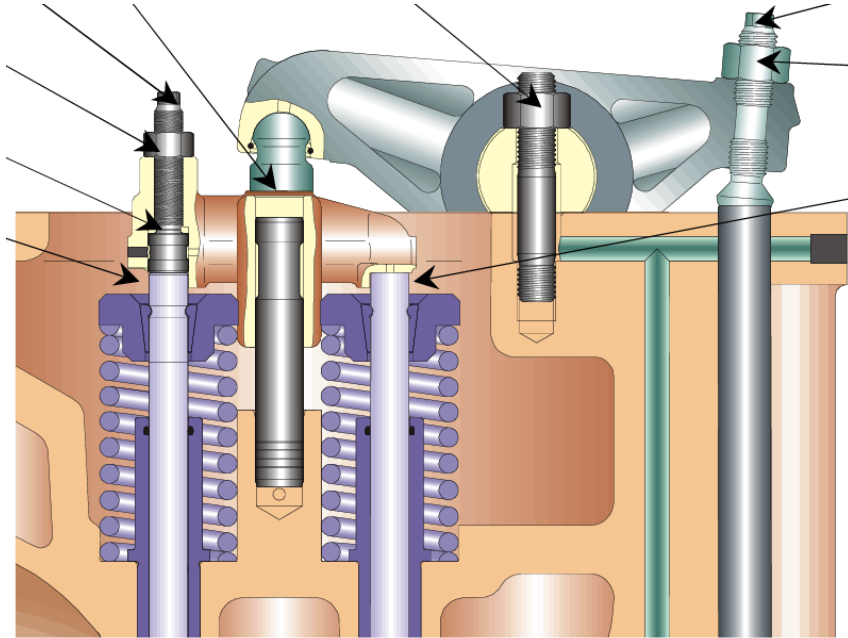
Fig.1 The cooling water flow in cylinder head (실린더 헤드내의 냉각수의 흐름)



Part List

Item No.	Description	Q'ty / Cyl.	Weight	Remarks / See Note
310	<u>Rocker arm complete - intake</u>	1	2.63	
311	<u>Rocker arm body - intake</u>	1	2.38	
312	Adjusting screw	2	0.12	
313	Ball seat	2	0.09	
314	Nut	2		
315	Clip spring	2		
316	Fine threaded plug	4		
317	Bush for rocker arm	2		
320	<u>Rocker arm complete - exhaust</u>	1	2.63	
321	<u>Rocker arm body - exhaust</u>	1	2.38	
330	Support for rocker arm	1	2.3	
410	<u>Yoke - intake</u>	1	0.84 kg	
415	Adjusting screw - Yoke	2	0.07 kg	
416	Shaft pin for Yoke	2	0.26 kg	
419	Nut	2		
420	<u>Yoke - exhaust</u>	1	0.84 kg	
901	Stud for rocker arm support	2		
902	Nut	2		
907	Spacer for rocker arm support	2		

Intake & exhaust system



Cylinder Head

A21100

Water Jacket

A15100

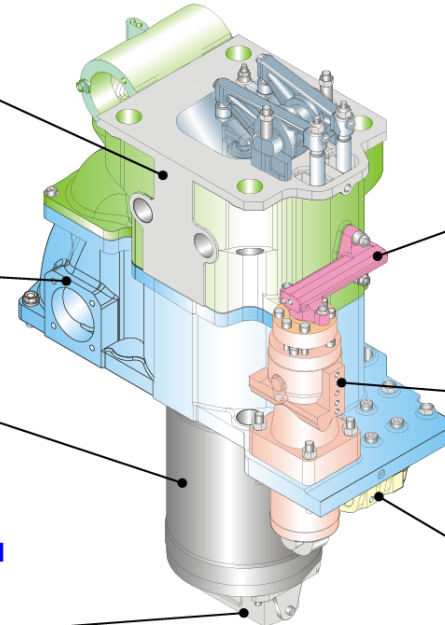
Cylinder Liner

A15100

Piston & Con Rod

A31100

A32000



High Pressure Block

A52300

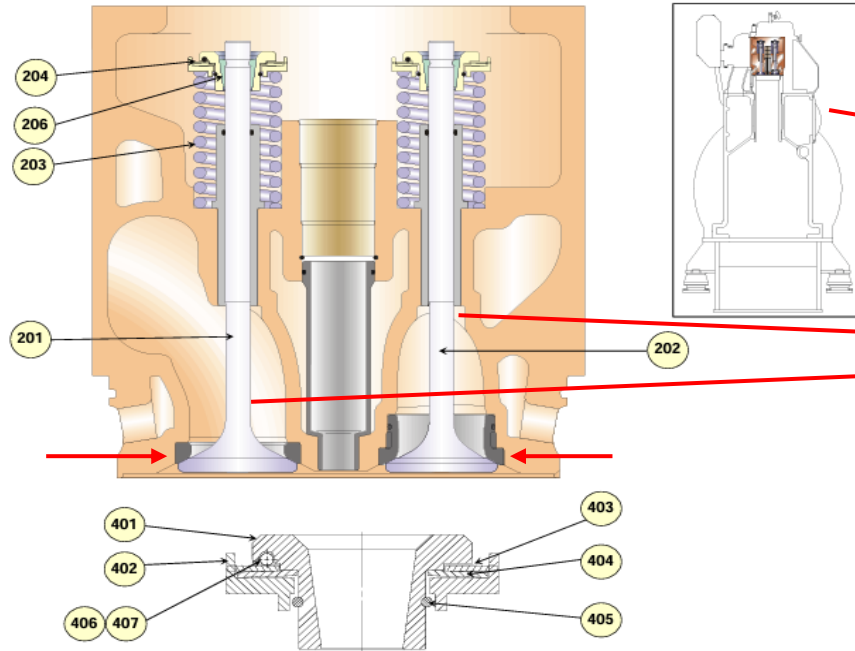
Fuel Injection Pump

A51000

Valve Train

A23000

Fig. 1 Cylinder Unit (실린더 유니트)



As shown in the left picture, when the engine is viewed from the side rather than the front,

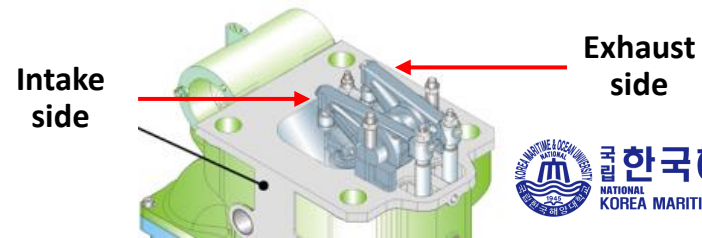
the valve spindle (intake) and valve spindle (exhaust) do not appear as shown in the left picture.

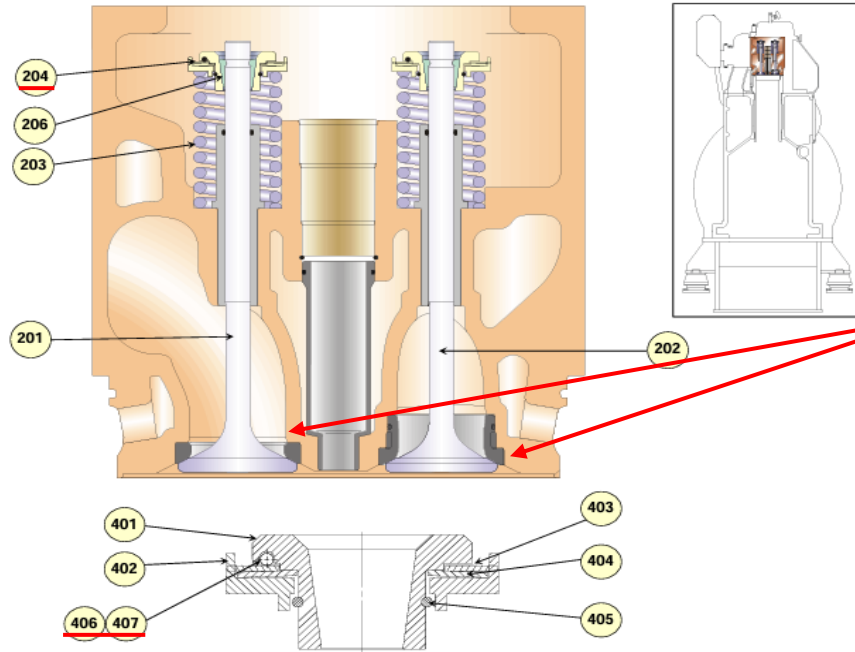
Instead, two valve spindles (intake) or two valve spindles (exhaust) appear together in one row.

The picture on the left shows the valve spindle (intake) and valve spindle (exhaust) together for understanding purposes only, and in reality, it is not like that.

Part List

Item No.	Description	Q'ty / Cyl.	Weight (kg)	Remarks / See Note
201	Inlet valve spindle	2	0.93	
202	Exhaust valve spindle	2	0.85	
203	Valve spring	4		
204	Roto cap	4	0.29	
206	Conical clamping piece	4	0.02	
401	Body	4		
402	Retainer	4		
403	Bearing seat	4		
404	Disc spring	4		
405	Stop ring	4		
406	Steel ball	24		
407	Coil spring	24		





The valve seats for the valve spindle (intake) and valve spindle (exhaust) are different because one side is where the pressurized air cooled by the air cooler enters, and the other side is where the hot exhaust gas is discharged.

The roto cap has a steel ball that allows the valve spindle to rotate.

Accordingly, the valve spindle and valve seat do not always have the same surface in contact, but instead, different surfaces always come into contact as they rotate.

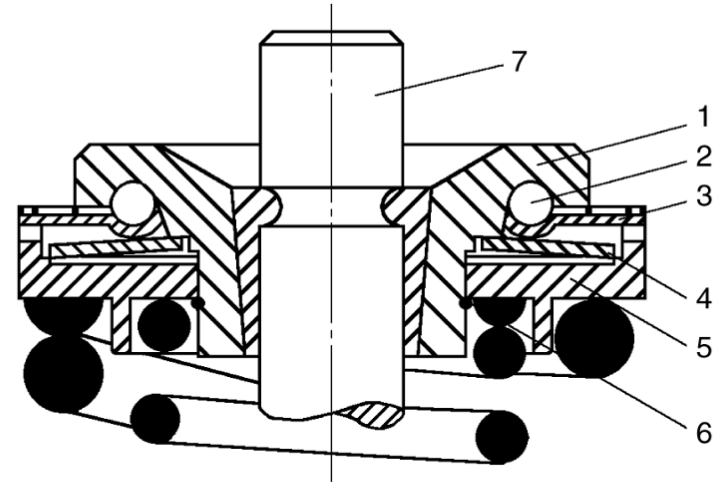
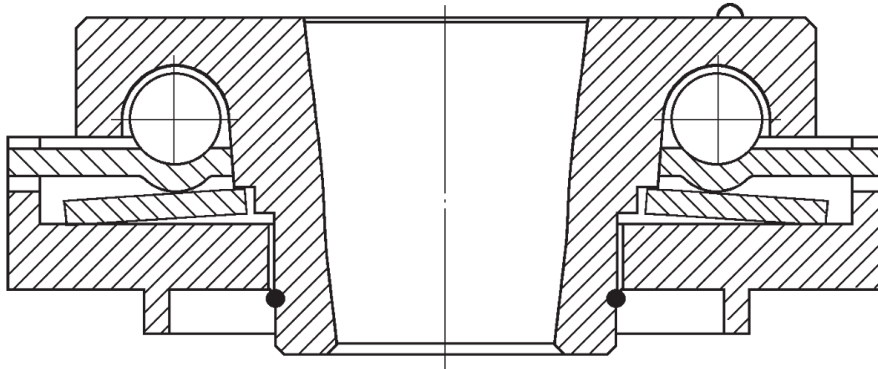
This has the effect of removing carbon deposits and preventing intake air or exhaust from leaking out by wearing out only one side.

Part List

Item No.	Description	Q'ty / Cyl.	Weight (kg)	Remarks / See Note
201	Inlet valve spindle	2	0.93	
202	Exhaust valve spindle	2	0.85	
203	Valve spring	4		
204	Roto cap	4	0.29	
206	Conical clamping piece	4	0.02	
401	Body	4		
402	Retainer	4		
403	Bearing seat	4		
404	Disc spring	4		
405	Stop ring	4		
406	Steel ball	24		
407	Coil spring	24		

Roto cap

<https://www.youtube.com/watch?v=uBS99Ee5Is0>



- | | | | |
|---|----------------|---|---------------|
| 1 | Retainer body | 2 | Balls |
| 3 | Ball race | 4 | Spring washer |
| 5 | Seating collar | 6 | Retainer ring |
| 7 | Valve spindle | | |

Roto cap



Roto cap

Rotocap-Working-Animation



Roto cap

ROTOCAP? OVERHAUL



Intake & exhaust system

Adjustment of valve clearance (Tappet clearance)

After adjustment of the yoke balancing, continue to adjust valve clearance as follows;

1. Check the wearing parts of rocker arm assembly maintained correctly.
2. Check the nut (5) for fixing rocker arm tightened correctly or tighten them again.
Tightening torque: 180 Nm (with molycote)
3. Loosen the lock nut (3) and adjusting screw (4) of rocker arm.
4. Insert a feeler gauge between rocker arm and yoke (D).
Thickness of feeler gage : 0.5 mm

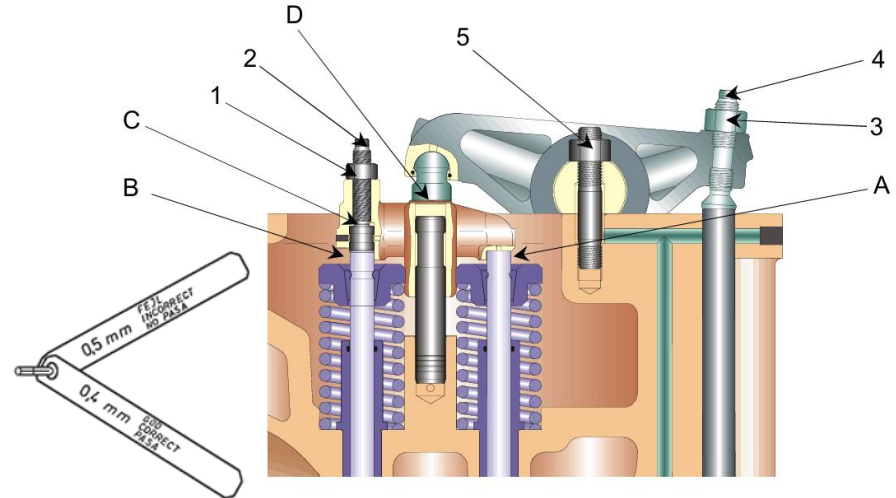
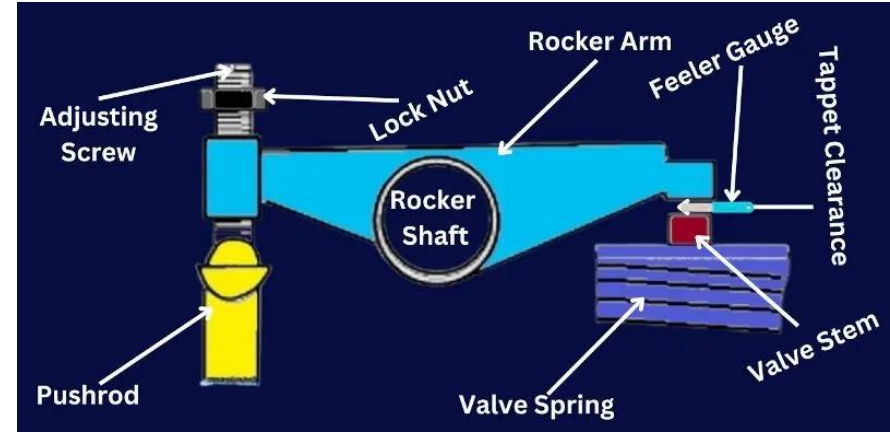


Fig. 1 Adjustment of valve clearance (밸브 틈세의 조정)

Intake & exhaust system

Adjustment of valve clearance (Tappet clearance)

5. Rotate the adjusting screw (4) until bottom of the adjusting contacting with push rod.
6. Tighten the lock nut (3) while holding the adjusting screw (4) not to be turned.
7. After locking, measure the clearances again for confirmation. If necessary, readjust.

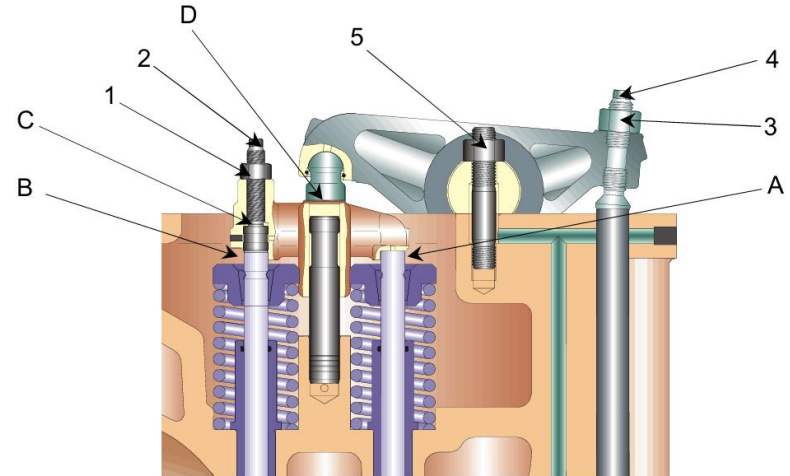
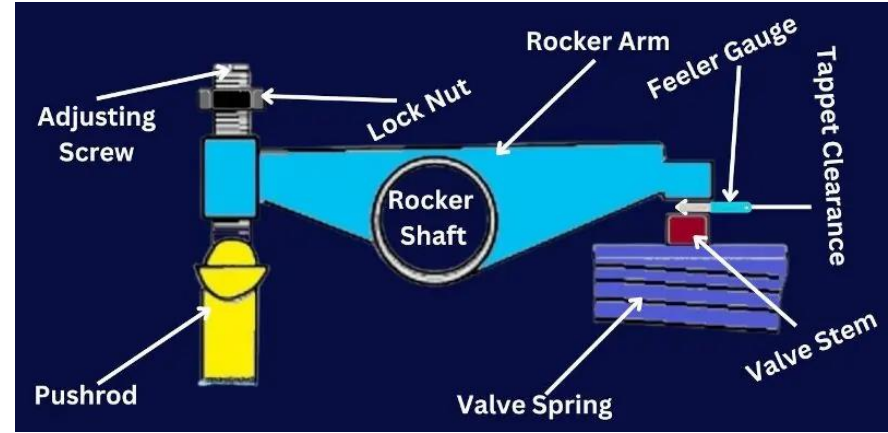


Fig. 1 Adjustment of valve clearance (밸브 틈세의 조정)

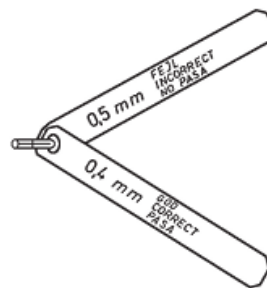
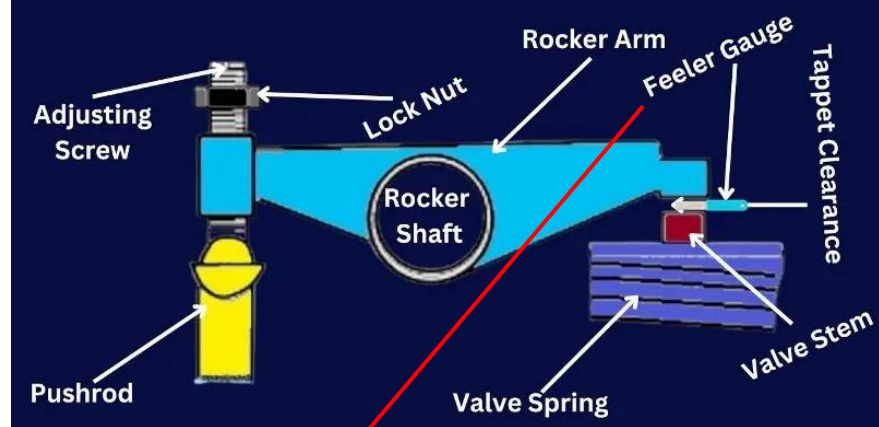
Intake & exhaust system

For HiMSEN engines, the feeler gauges on the intake and exhaust sides have the same thickness.

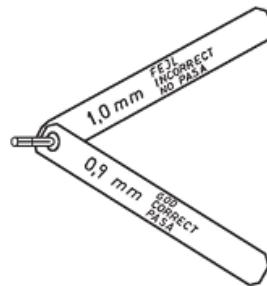
However, for MAN B&W engines, the feeler gauges on the intake and exhaust sides have different thicknesses.

The feeler gauges for checking the clearance have two gauges which are marked "incorrect" and "correct", the latter to be used when adjusting the valve clearance.

After tightening up the counter nuts on rocker arms and valves bridge, be sure that the feeler gauges marked "correct" can be inserted into the two clearances simultaneously as where it must not be possible to insert the gauges marked "incorrect".



Intake side



Exhaust side



한국해양대학교
NATIONAL KOREA MARITIME & OCEAN UNIVERSITY

Intake & exhaust system

Adjustment for balancing yoke

As two intake or exhaust valves are pressed by one yoke simultaneously, the yoke should be well balanced by precise adjustment as follows.

1. Rotate crankshaft so that the valves are in fully closed position. The corresponding cylinder needs to be at around firing TDC.
2. Check both push rods are free from spring forces and have clearances.
3. Loosen the lock nuts (1, 3) and adjusting screws (2, 4).
4. Press the non-adjusting side of yoke in order to contact with top of valve stem (A) firmly as shown on fig 1.

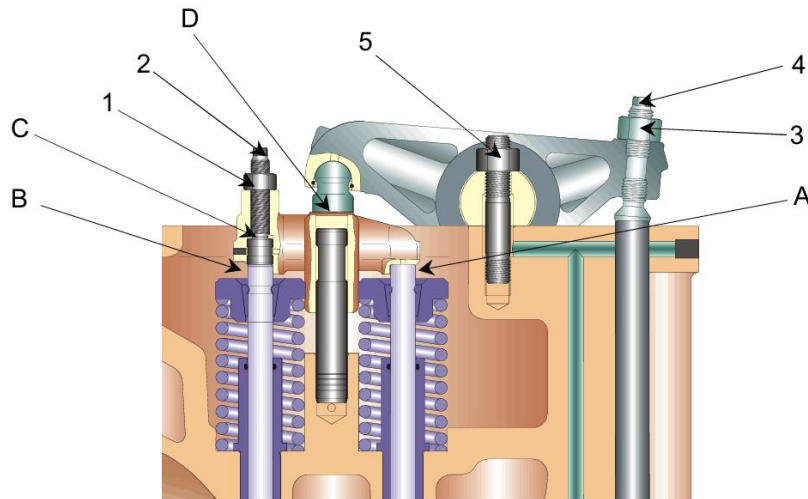


Fig. 1 Adjustment of valve clearance (밸브 틈새의 조정)

Intake & exhaust system

Adjustment for balancing yoke

5. While contacting (A) by pressing, rotate the adjusting screw (2) until contacting with top of the valve stem (B).
6. Press (A) side of the yoke and measure the clearance of (B) by feeler gauge.
7. Similarly, press (B) side of the yoke and measure the clearance of (A) by feeler gauge.
8. If the measured clearances are different, then unscrew the adjusting screw slightly and measure the clearances in turn by pressing the other side of the yoke.
9. Repeat the adjustment until the both clearances are same value, and then tighten the lock nut (1) while holding the adjusting screw (2) not to be turned.
10. After locking, measure the both clearances again for confirmation.

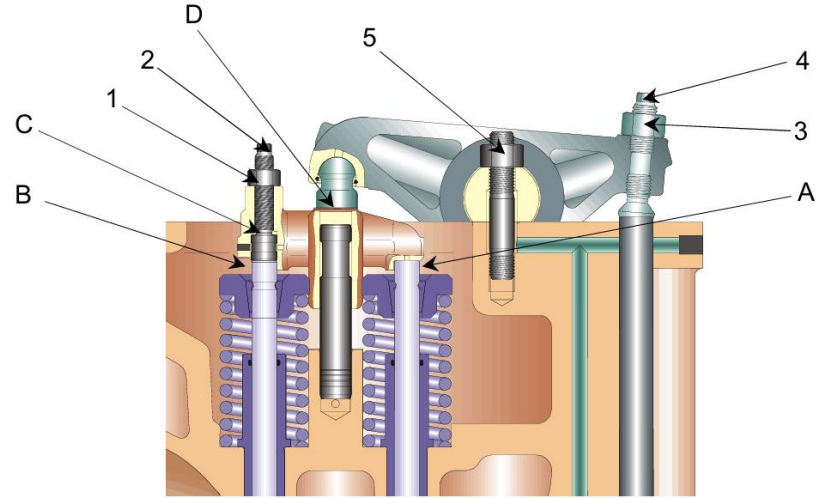


Fig. 1 Adjustment of valve clearance (밸브 틈새의 조정)

Intake & exhaust system

Adjustment for balancing yoke

Maladjustment of yoke clearance causes unbalance force on the yoke which results abnormal wear or damage on the yoke and guide pin.

If the guide pin or the yoke is damaged or broken, repeated unbalanced motion of the valve spindles may lead to failure of the valve spindle and serious damage on the engine.

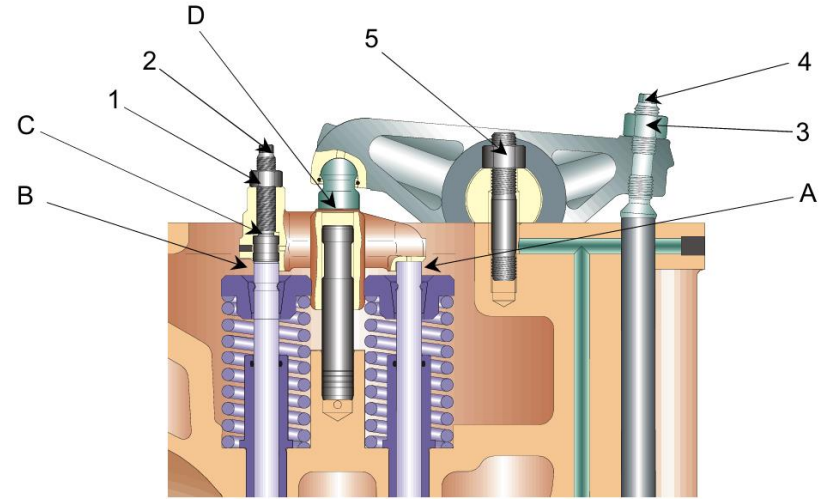
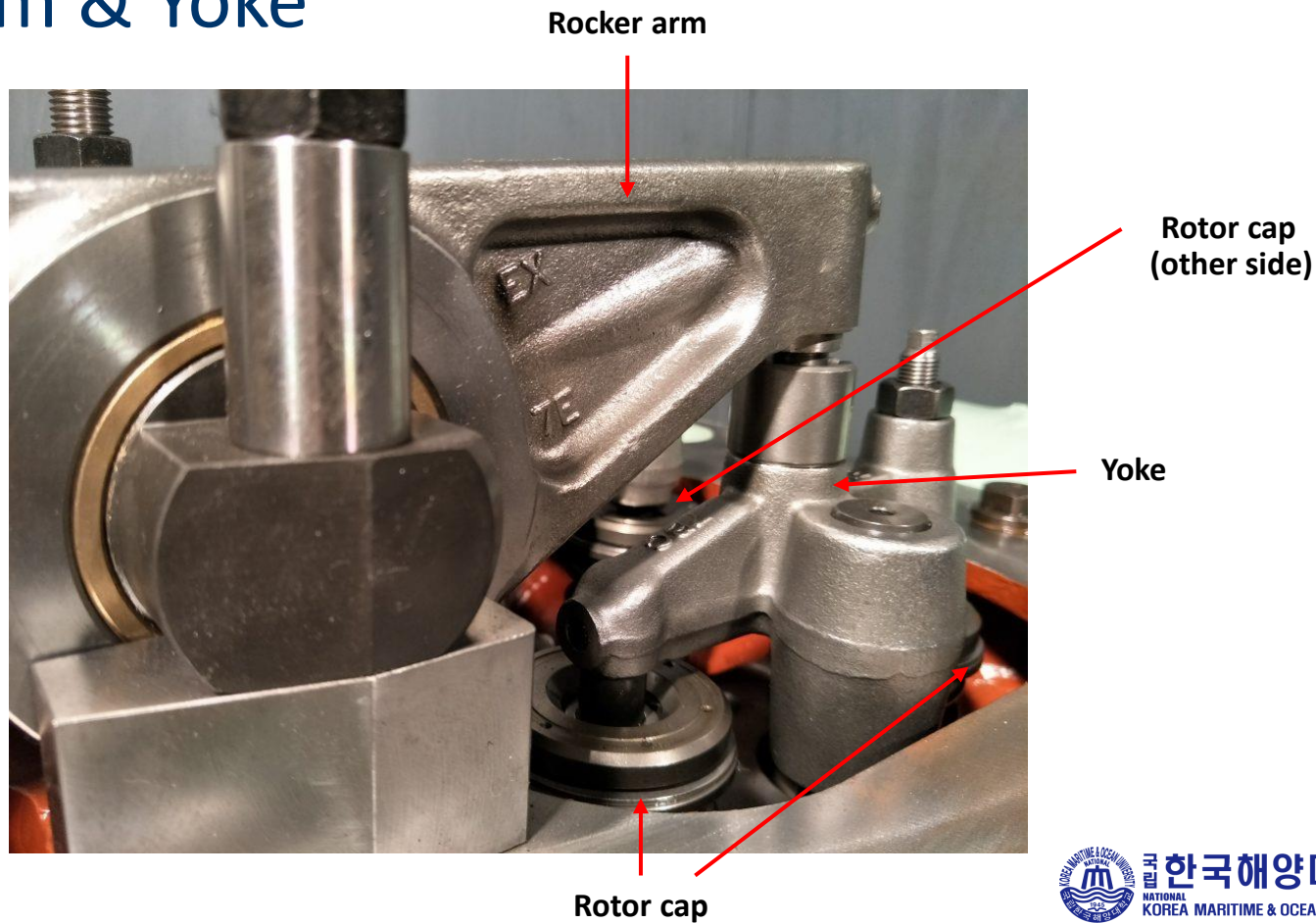


Fig. 1 Adjustment of valve clearance (밸브 틈새의 조정)

Rocker arm & Yoke



Tappet clearance



Tappet clearance



Tapet clearance



Tappet clearance



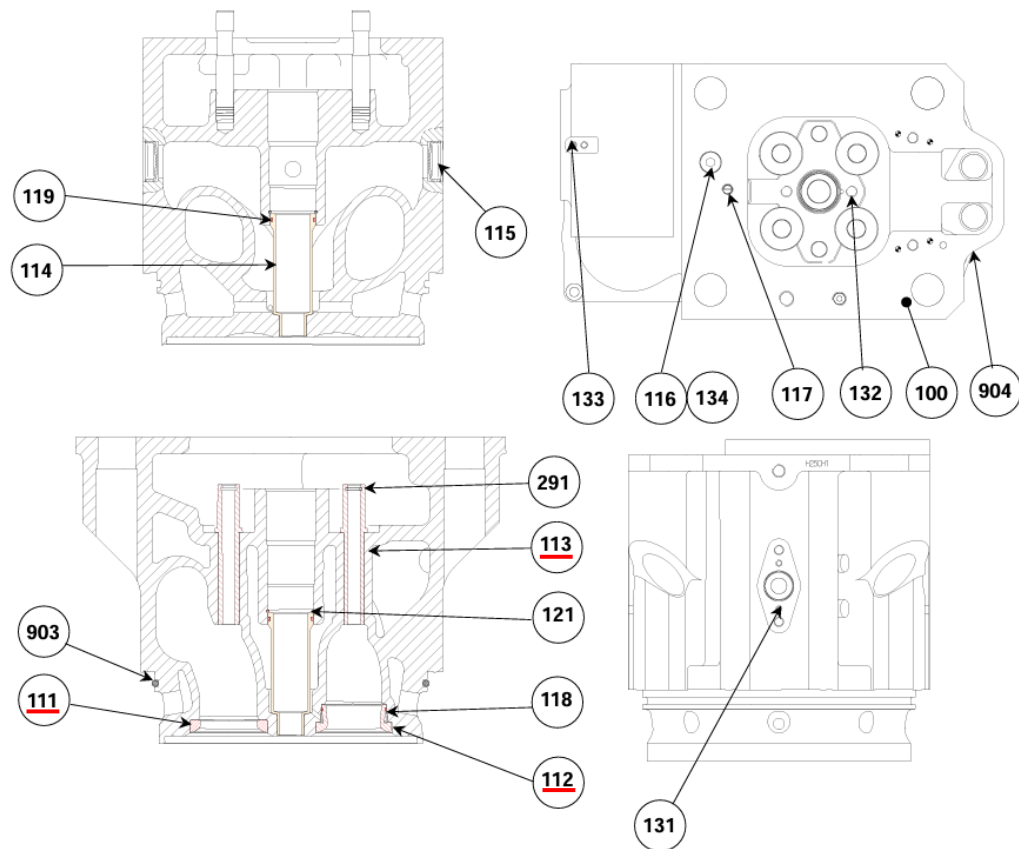
Tappet clearance



Tappet clearance



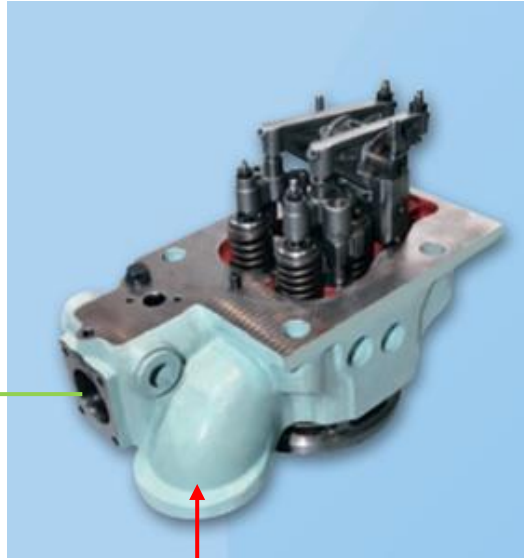
Cylinder head



Part List

Item No.	Description	Q'ty / Cyl.	Weight (kg)	Remarks / See Note
100	Cylinder head complete(water tight.)			Item no. 111 ~ 904
<u>111</u>	<u>Valve insert - inlet</u>	2	0.23	
<u>112</u>	<u>Valve insert - exhaust</u>	2	0.40	
<u>113</u>	<u>Valve guide</u>	4	0.35	
114	Bush for fuel valve	1	0.42	
115	Closing cover	4		
116	Plug	1		
117	Set screw	1		
118	O-ring	2		
119	O-ring	1		
121	Locking ring	1		
131	Set screw	2		Only for N.C.O system
132	Stud bolt	1		
133	Set screw	1		
134	Gasket	1		
291	O-ring	4		
903	O-ring	1		
904	Set screw	1		

Cylinder head

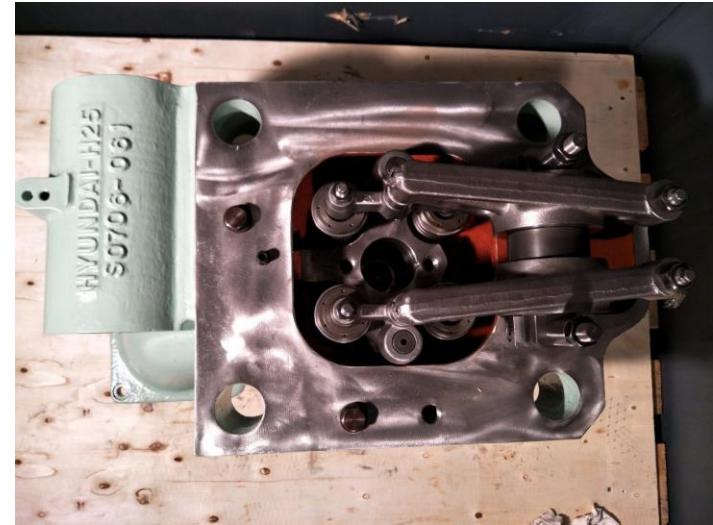
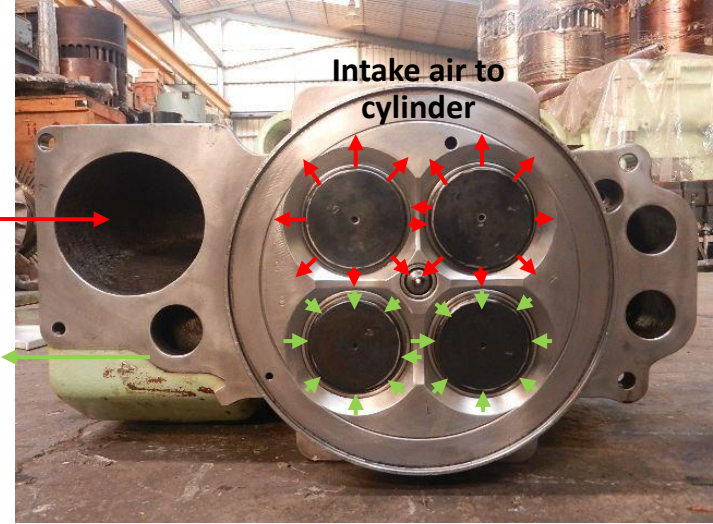


Exhaust gas to
exhaust manifold

Intake air from air cooler

Intake air from
air cooler

Exhaust gas to
exhaust manifold



Cylinder head (MAN B&W)

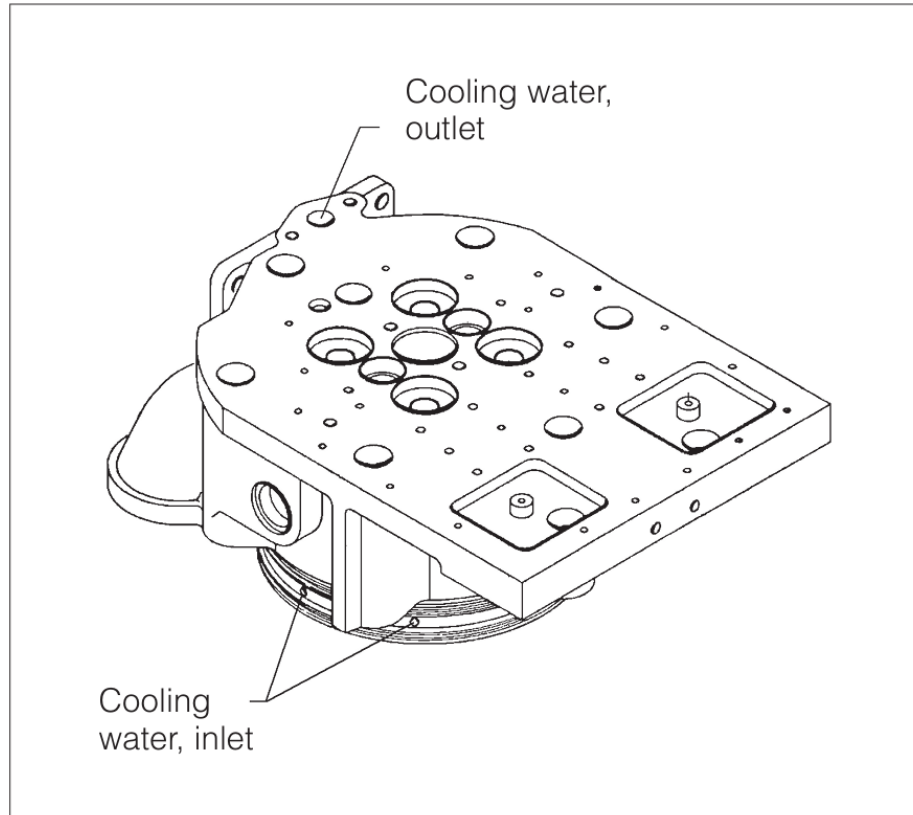
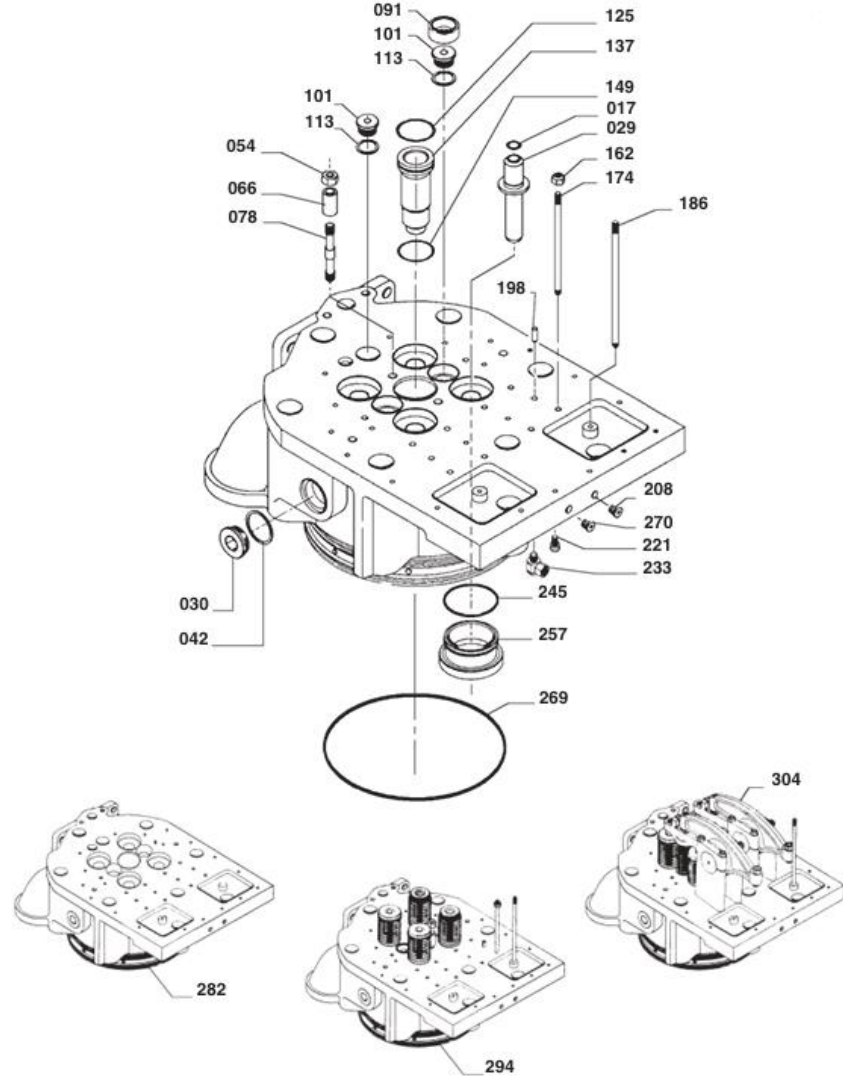


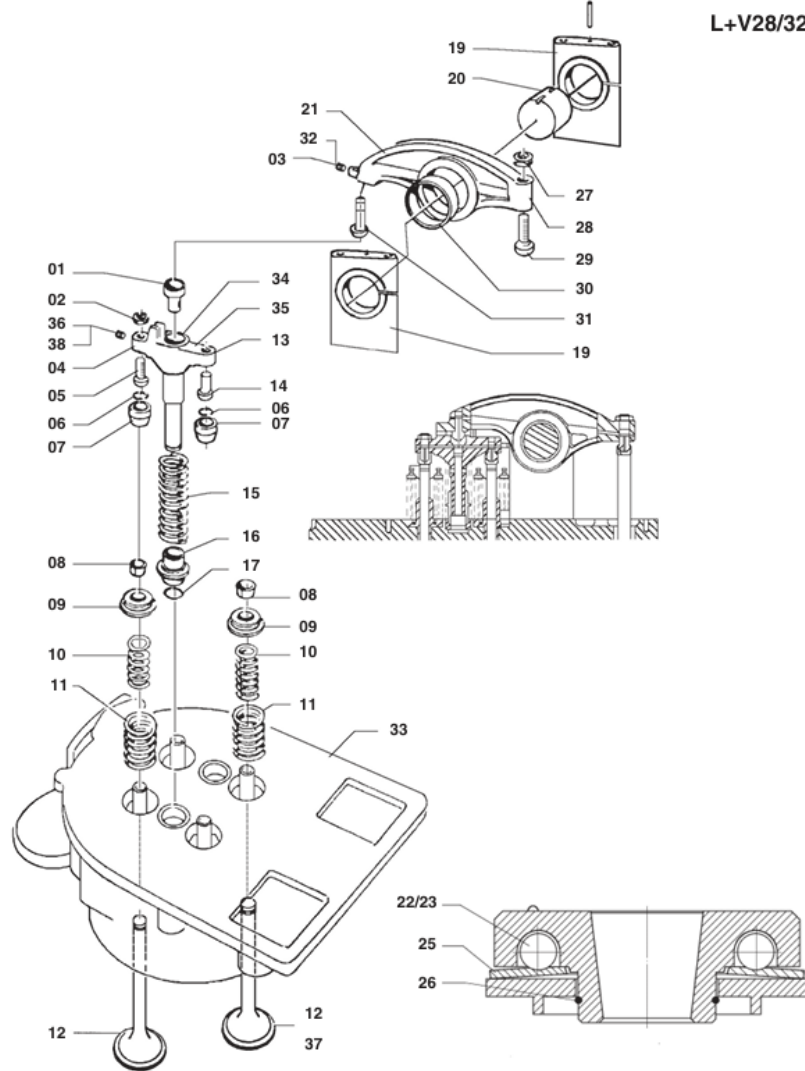
Fig. 1. Cylinder Head



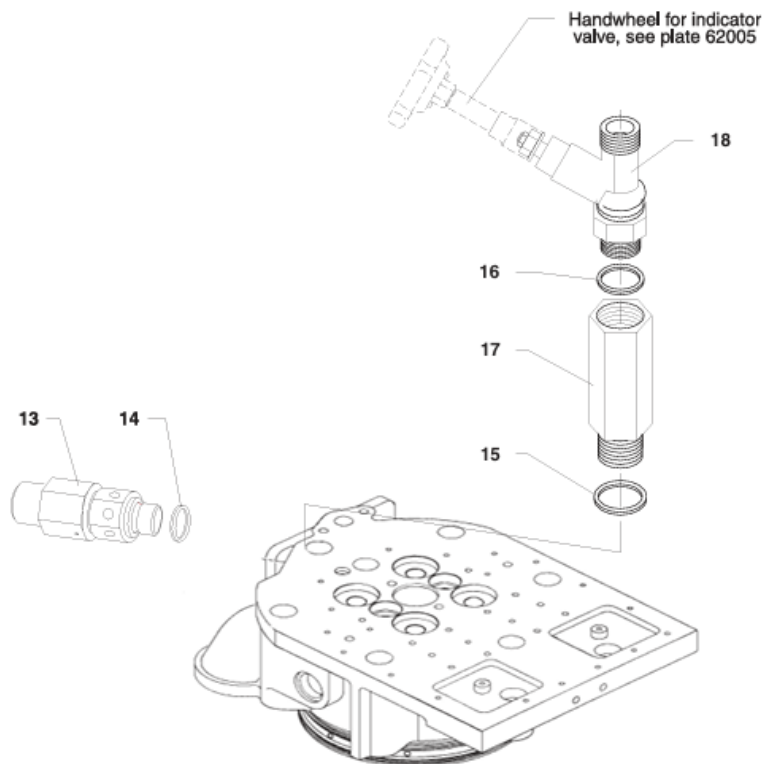
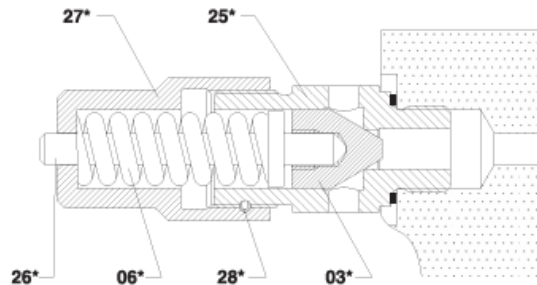
Item No.	Qty.	Designation	Benævnelse
017	4/C	O-ring	O-ring
029	4/C	Valve guide, inlet and exhaust	Ventilstyr, indstr. og udstr.
030	6/C	Plug screw	Propskrue
042	6/C	Gasket	Pakning
054	2/C	Nut	Møtrik
066	2/C	Distance pipe	Afstandsør
078	2/C	Stud	Tap
091	2/C	Thrust collar	Trykflange
101	4/C	Plug screw	Propskrue
113	4/C	Gasket	Pakning
125	1/C	Snap ring	Låsering
137	1/C	Sleeve for fuel valve	Foring for br. ventil
149	1/C	O-ring	O-ring
162	8/C	Nut	Møtrik
174	8/C	Stud	Tap
186	4/C	Stud	Tap
198	4/C	Spring pin	Fjederstift
208	2/C	Plug screw	Propskrue
221	2/C	Screw	Skrue
233	2/C	Coupling	Kobling
245	4/C	O-ring	O-ring
257	4/C	Valve seat ring, (inlet and exhaust)	Ventilsædering, (indstr. og udstødning)
269	2/C	O-ring	O-ring
270	2/C	Plug screw	Propskrue
282		Cylinder head with valve guide sleeve for fuel valve and valve seat rings	Cylinderdæksel monteret med spindelstyr, foring for br. ventil og ventilsæderinge
294		Cylinder head (as 282) with valve spindles installed	Cylinderdæksel (som 282), monteret med ventilspindler

Item No.	Qty.	Designation	Benævnelse
304		Cylinder head (as 294) with brackets and rocker arms installed	Cylinderdæksel (som 294) monteret med bukke og vippearne.

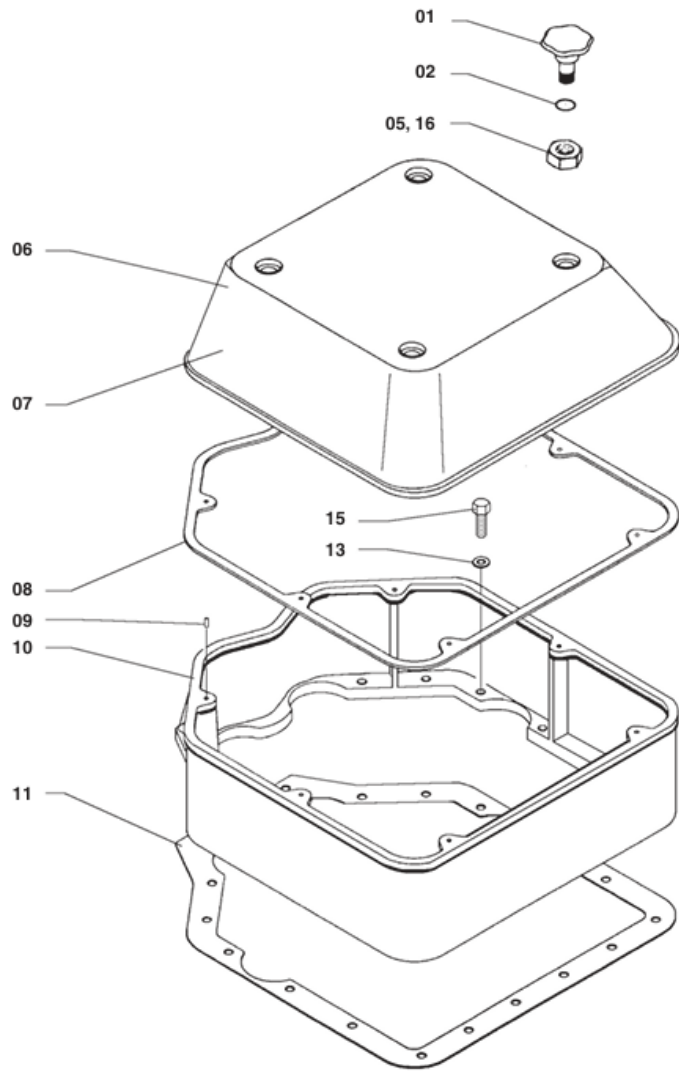
L+V28/32H



Item No.	Qty.	Designation	Benævnelse	Item No.	Qty.	Designation	Benævnelse
01	2/C	Thrust piece	Trykstykke	28	2/C	Rocker arm	Vippearm
02	2/C	Nut	Motrik	29	2/C	Thrust screw	Trykskrue
03	4/C	Plug screw	Propskruer	30	2/C	Bearing bush	Løjebøsning
04	1/C	Valve bridge, inlet	Ventilbro, indstrømning	31	2/C	Thrust pin	Trykskrue
05	2/C	Thrust screw	Trykskrue	32	/I	Loctite 572	Loctite 572
06	4/C	Circlip	Fjederring	33	1/C	Cylinder head as Plate 60501 + Plate 60502	Cylinderdæksel som Plate 60501 + Plate 60502
07	4/C	Thrust piece	Trykstykke	34	1/C	Valve bridge, inlet complete incl. item 01, 02, 04, 05, 06, 07, 14, 15, 16 and 17	Ventilbro, indstrømning komplet inkl. item 01, 02, 04, 05, 06, 07, 14, 15, 16 og 17
08	4/C	Conical ring in 2/2	Konisk ring 2/2	35	1/C	Valve bridge, exhaust complete incl. item 01, 02, 05, 06, 07, 13, 14, 15, 16 and 17	Ventilbro, udstødning komplet inkl. item 01, 02, 05, 06, 07, 13, 14, 15, 16 og 17
09	4/C	Rotacap, complete incl. item 22, 23, 25, 26	Rotationsgiver, komplet inkl. item 22, 23, 25, 26	36	4/C	Screw	Skrue
10	4/C	Inner spring	Indvendig fjeder	37	2/C	Valve spindle, exhaust (only for stationary engines)	Ventilspindel, uds. (kun for stationære motorer)
11	4/C	Outer spring	Udvendig fjeder	38	/I	Loctite 243	Loctite 243
12	4/C	Valve spindle, inlet and exhaust (valve spindle exh. only for marine engines)	Ventilspindel, inds. og uds. (ventilspindel uds. kun for marine motorer)				
13	1/C	Valve bridge, exhaust	Ventilbro, udstødning				
14	2/C	Thrust piece	Trykstykke				
15	2/C	Spring	Fjeder				
16	2/C	Ball guide	Kuglestyr				
17	2/C	Securing ring	Sikring				
19	4/C	Rocker arm bracket incl spring pin	Buk for vippearm inkl spændstift				
20	2/C	Rocker arm shaft	Aksel for vippearm				
21	2/C	Rocker arm, compl. (item 03, 27, 28, 29, 30, 31)	Vippearm, komplet (item 03, 27, 28, 29, 30, 31)				
22	24/C	Spring	Fjeder				
23	24/C	Ball	Kugle				
25	4/C	Spring washer	Fjederskive				
26	4/C	Retainer ring	Låsering				
27	2/C	Nut	Motrik				



Item No.	Qty.	Designation	Benævnelse
03*	1/C	Valve cone	Ventilkugle
06*	1/C	Pressure spring	Trykfjeder
13	1/C	Safety valve, complete incl. item 03, 06, 25, 26, 27, 28	Sikkerhedsventil, komplet inkl. item 03, 06, 25, 26, 27, 28
14	1/C	Gasket	Pakning
15	1/C	Gasket	Pakning
16	1/C	Gasket	Pakning
17	1/C	Connecting piece	Forbindelsesstykke
18	1/C	Indicator valve, complete	Indikatorventil, komplet
25*	1/C	Valve housing	Ventilhus
26*	1/C	Valve spindle	Ventilspindel
27*	1/C	Cap nut	Dækselmøtrik
28*	1/C	Split pin	Fjedernot



Item No.	Qty.	Designation	Benævnelse
01	4/C	Handle	Håndtag
02	4/C	O-ring	O-ring
05	4/C	Nut	Møtrik
06	1/C	Top cover	Topdæksel
07	1/C	Top cover, compl. incl. item Nos. 01, 02, 05, 06	Topdæksel, kompl. inkl. pos. nr. 01, 02, 05, 06
08	1/C	Gasket	Pakning
09	6/C	Spring pin	Fjederstift
10	1/C	Coaming	Karm
11	1/C	Gasket	Pakning
13	18/C	Washer	Skive
15	18/C	Screw	Skrue
16	/I	Loctite 638	Loctite 638



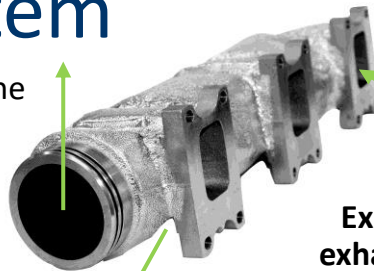
Intake & exhaust system

The exhaust gases are directed to the turbine of the turbocharger.

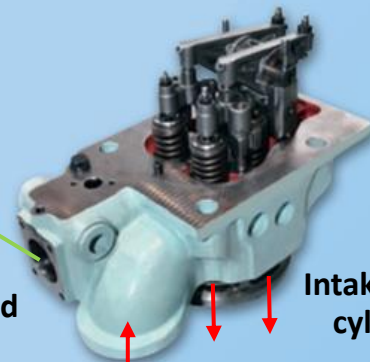
This is for reference only, the actual ship's exhaust manifold looks slightly different.

The exhaust manifold is inside the cover.

Exhaust manifold

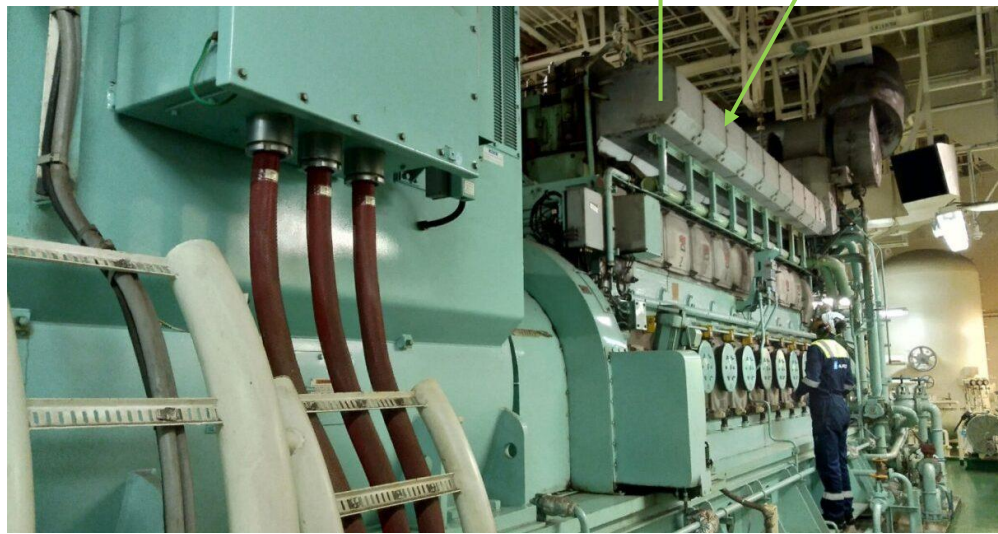


Exhaust gas to
exhaust manifold

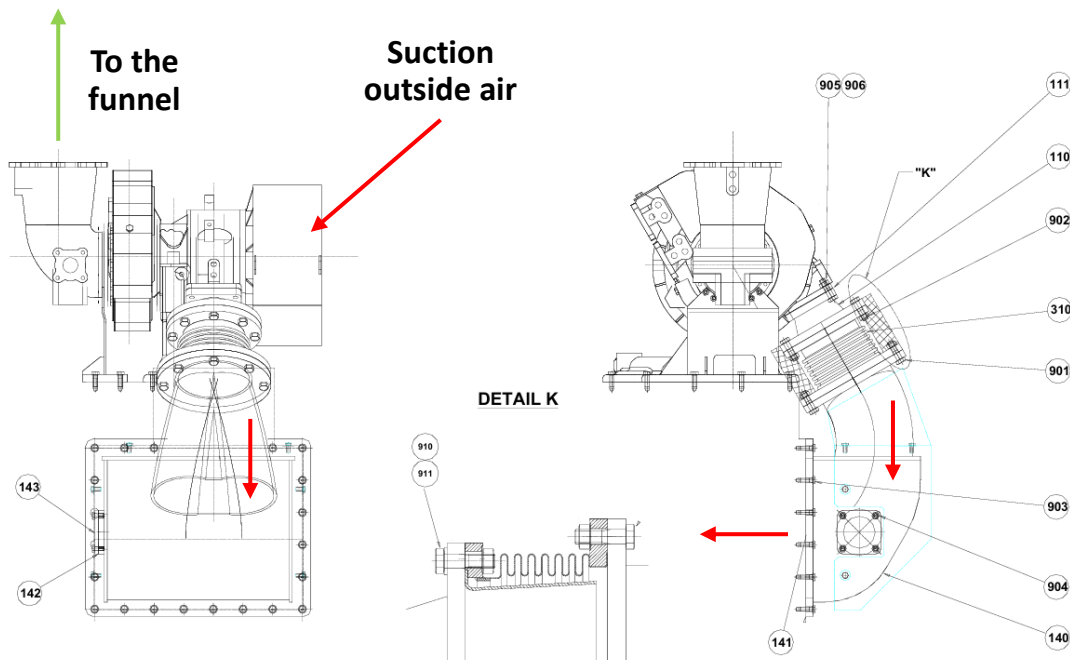


Intake air to
cylinder

Intake air
from air
cooler



Intake & exhaust system

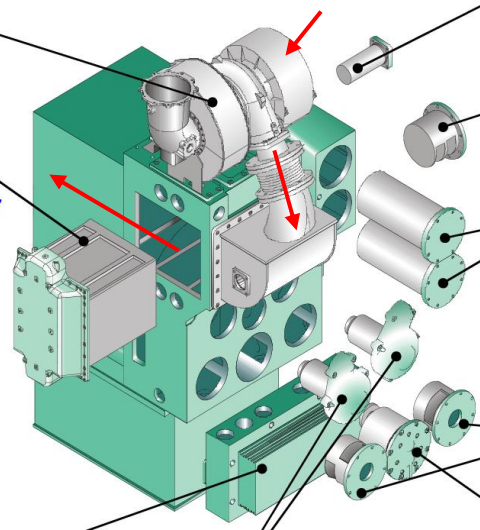


T/C Manual

Turbocharger

A84000

Charger Air Cooler



Part List (for 5H21/32)

Item No.	Description	Q'ty /Eng.	Weight (kg)	Remarks / See Note
000	Intake air duct ass'y	1		
110	Interpiece diffuser	1		
111	Gasket for compressor outlet	1		
140	Air inlet box	1		
141	Gasket	1		
142	Gasket	1		
143	Cover	1		
310	Axial compensator	1		
901	Bolt (M16x60).	16		
902	Nut (M16)	16		
903	Bolt (M12x35)	24		
904	Bolt (M12x25)	4		
905	Bolt (M12x45)	4		
906	Nut (M12)	4		
910	Bolt(M16x55)	8		
911	Washer	8		

Intake & exhaust system

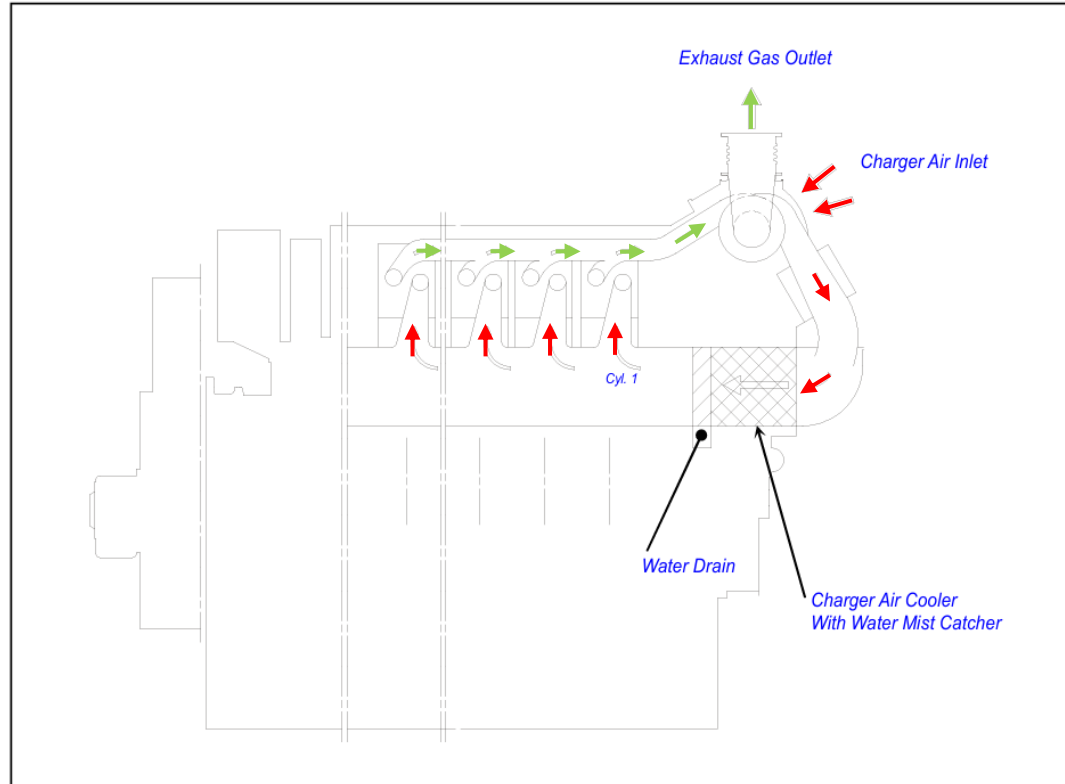
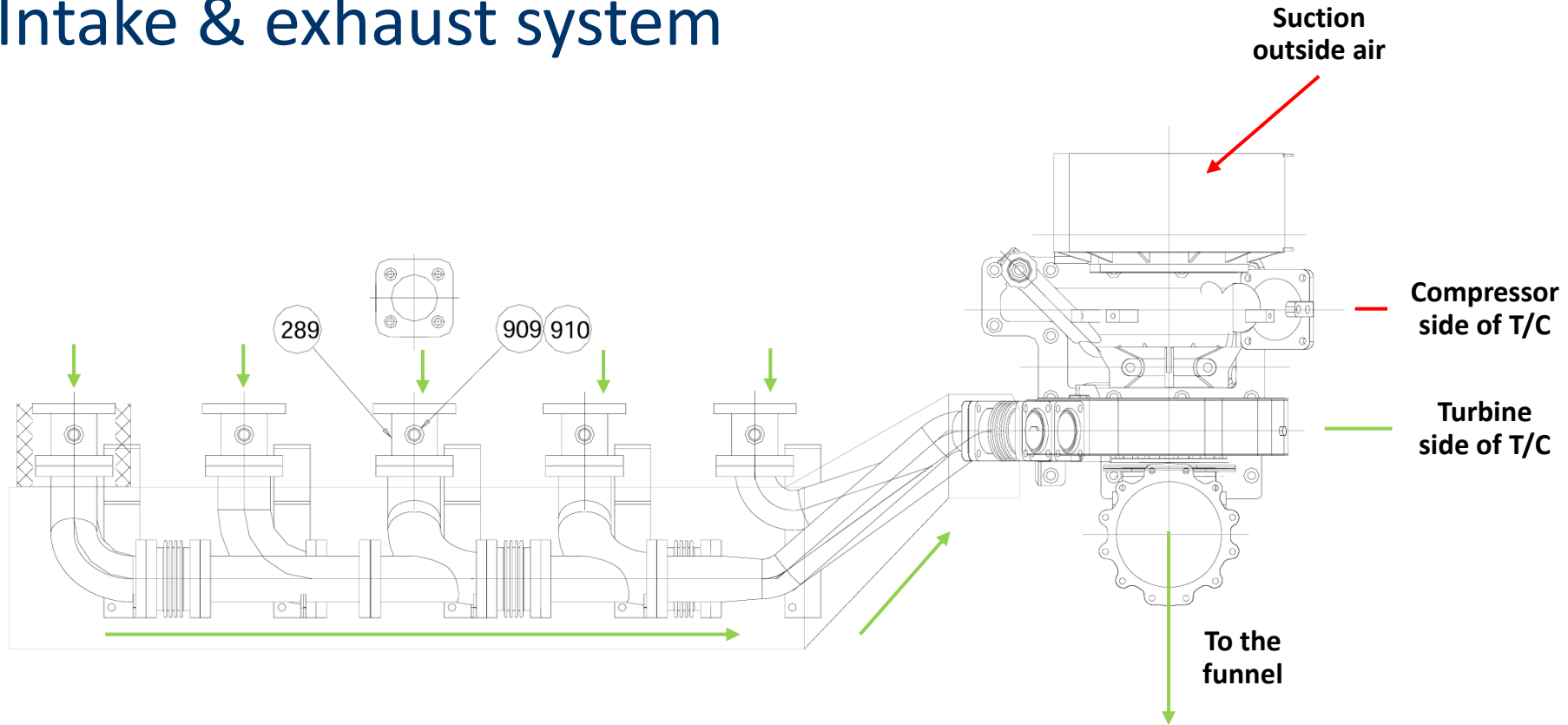
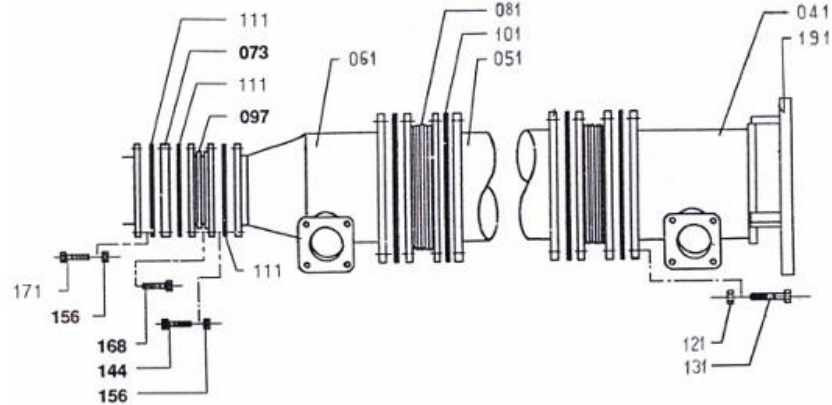


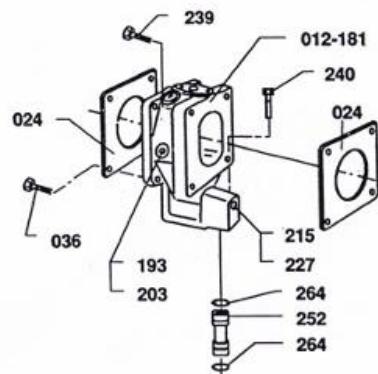
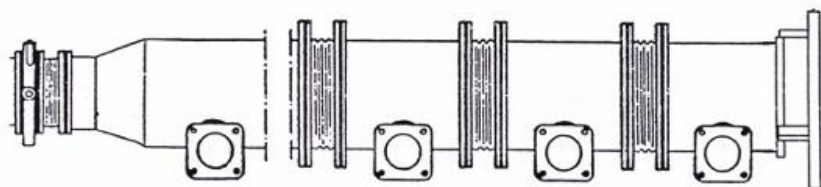
Fig.1 Diagram for combustion air and exhaust gas system (공기 및 배기 개스 시스템 선도)

Intake & exhaust system



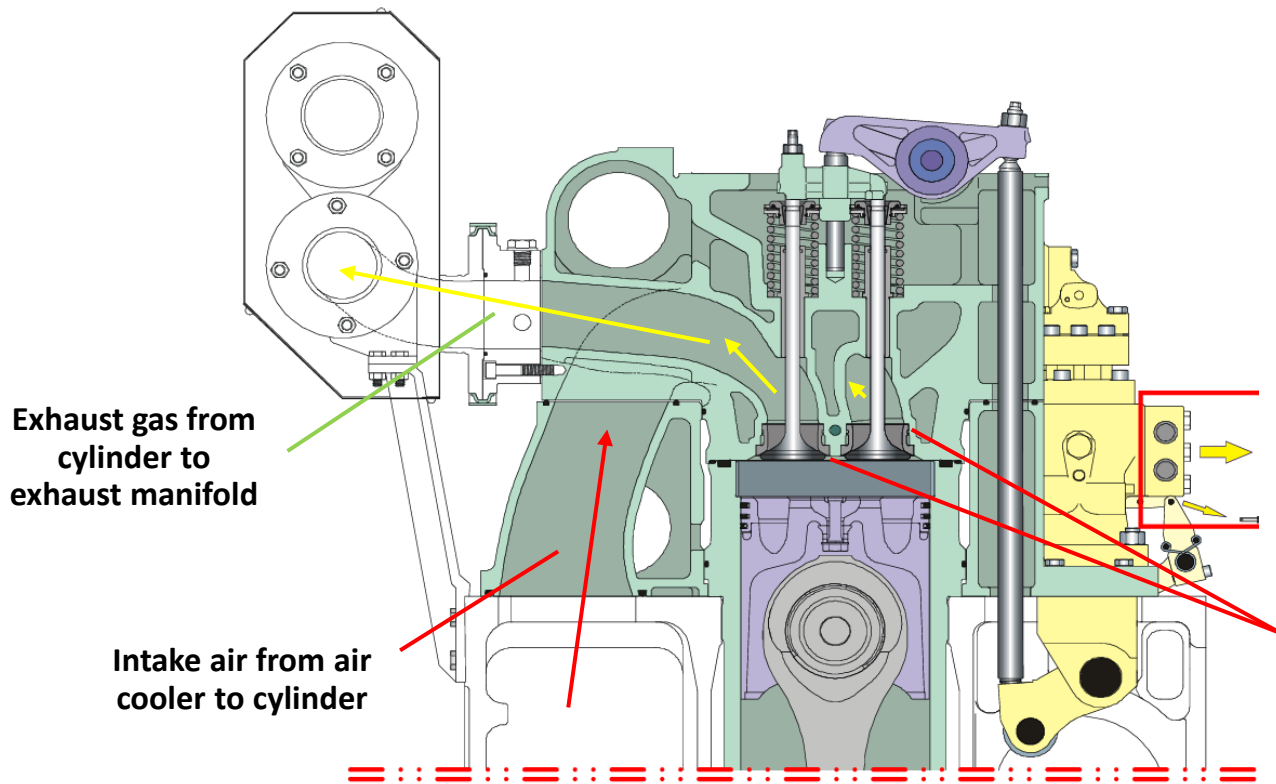


Exhaust manifold



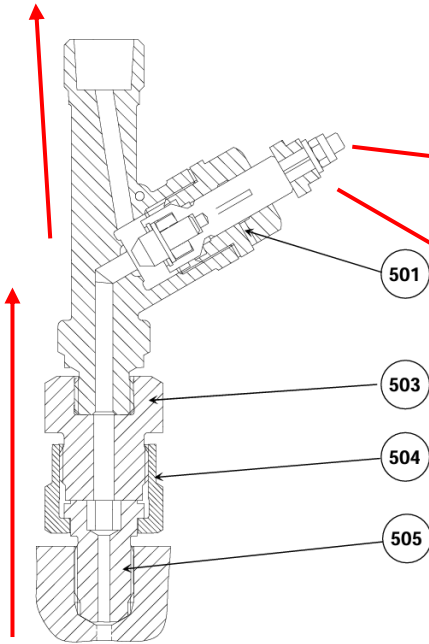
Item No.	Qty.	Designation	Item No.	Qty.	Designation
012	1/C	Intermediate piece, complete, incl. item 181, 193, 203, 215 and 227	193	3/C	Plug screw
024	2/C	Gasket	203	3/C	Gasket
036	4/C	Screw	215	1/C	Plug screw
041	1/E	Exhaust pipe	227	1/C	Sealing ring
051		Exhaust pipe	239	4/C	Screw
	3/E	5 cyl. engine	240	1/C	Screw
	4/E	6 cyl. engine	252	1/C	Cooling water nipple
061	1/E	Exhaust pipe for cyl. 1	264	2/C	O-ring
073	1/E	Flange			
081		Compensator			
	4/E	5 cyl. engine			
	5/E	6 cyl. engine			
097	1/E	Compensator			
101		Gasket			
	8/E	5 cyl. engine			
	10/E	6 cyl. engine			
111	3/E	Gasket			
121		Nut			
	96/E	5 cyl. engine			
	120/E	6 cyl. engine			
131		Screw			
	96/E	5 cyl. engine			
	120/E	6 cyl. engine			
144	8/E	Screw			
156	16/E	Nut			
168	8/E	Screw			
171	8/E	Screw			
181	1/C	Intermediate piece			
191	1/E	End cover, aft			

Intake & exhaust system



If you look at the valve seat, you can see that the part shown in the picture is the exhaust side.

Test cock (indicator valve) of cylinder head



Part List

Item No.	Description	Q'ty / Cyl.
501	Indicator valve (M20xP=2)	1
503	Connection socket	1
504	Union	1
505	Threaded socket	1

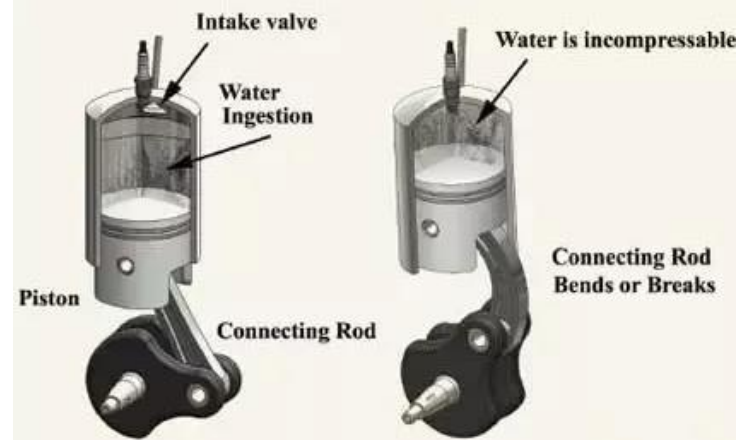


Test cock of cylinder head

The test cock is used to determine whether there is any leakage of cooling water, lubricant, or oil inside the cylinder due to the reciprocating movement of the piston by only air starting while it is open at the initial start of the G/E. Failure to check this can result in hydrolock, which can lead to major problems.

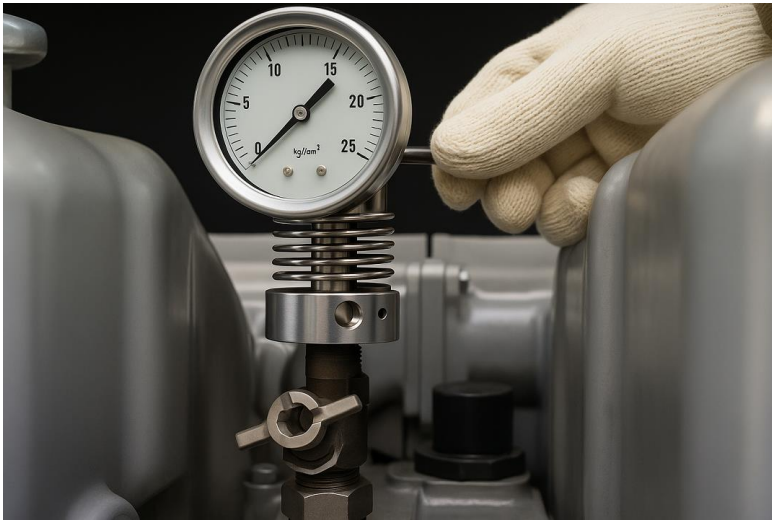
This process is called “Air blowing” on ships.

Air blow also prevents any explosive gases remaining in the cylinder from igniting during the compression process, causing pre-ignition.



Test cock of cylinder head

During G/E operation, a pressure indicator can be installed on the test cock to measure Compression Pressure (P_{comp}) and Maximum Combustion Pressure (P_{max}).



Measurement of P_{comp} and P_{max}



Measurement of P_{comp} and P_{max}

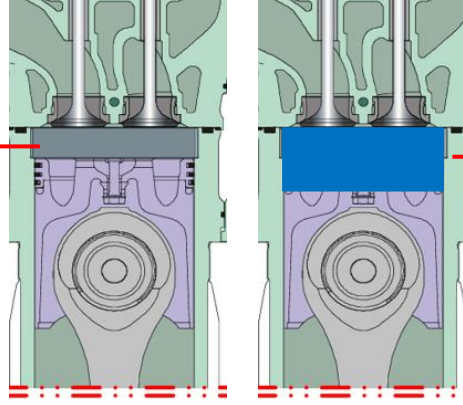


Hydrolock

Hydrolock (a shorthand notation for hydrostatic lock or hydraulic lock) is an abnormal condition of any device which is designed to compress a gas by mechanically restraining it caused by a liquid entering the device.

In the case of a reciprocating internal combustion engine, a piston cannot complete its travel and mechanical failure may occur if a volume of liquid greater than the volume of the cylinder at its minimum (end of the piston's stroke) enters the cylinder, due to the incompressibility of liquids.

Empty space



Imagine that the volume of cooling water is greater than the empty space.

In this state, when the piston reciprocates, the connecting rod is bent due to liquid compression, as shown in the figure below.



References

- 1) <https://www.mdpi.com/2504-186X/5/2/11>
- 2) <https://www.liverpoolecho.co.uk/news/liverpool-news/ship-bosses-say-huge-plumes-14027331>
- 3) <https://pi-dubai.com/diesel-engine-carbon-buildup-the-hidden-threat-to-efficiency-and-longevity/>
- 4) https://commons.wikimedia.org/wiki/File:Diesel_generator_on_an_oil_tanker.jpg
- 5) <http://ayriko.com/generator-engine-cylinder-head/>
- 6) <https://www.indiamart.com/proddetail/diesel-generator-cylinder-head-2852950365130.html?srsltid=AfmBOooD17bRGA4uyBhdxT3DydluqVU6t6YfFHESXMoGXKP49DFmKD00>
- 7) <https://www.mdps.nl/hyundai-himsen/>
- 8) https://www.linkedin.com/posts/kuberpada_hyundai-himsen-cylinderhead-activity-7074265985796517890-Muqa/



References

- 9) <https://www.tradewheel.com/p/himsen-ship-marine-diesel-engine-korea-1336131/>
- 10) https://www.indiamart.com/proddetail/himsen-h21-32-major-spares-crankshaft-block-cylinder-head-liner-piston-conrod-25450472488.html?srltid=AfmBOoplrtlwJDdRnXYgPP2w68nd_EjsQyhJxJam88eH59q8uXrC7AwB
- 11) <https://www.brechmann-guss.de/insulation-echaust-manifold-ni-resist?lang=en/>
- 12) <https://marineinfo.com/auxiliary-engine-fuel-oil-supply-system/>
- 13) https://www.youtube.com/watch?v=8gUBI7F5n4s&list=PLXNEJpAaCDcwZZ0Ym_fN9G7wDS74UquMb&index=13

References

- 14) <https://maritimeducation.com/how-to-check-performance-data-in-marine-diesel-engines-pmax-measurement-guide/>
- 15) <https://en.wikipedia.org/wiki/Hydrolock>
- 16) https://www.youtube.com/watch?v=eJcSvaW_e4w
- 17) <https://www.youtube.com/watch?v=40o21uleNDI>
- 18) <https://www.youtube.com/watch?v=893q7mYklkU>
- 19) <https://www.youtube.com/watch?v=vfTlkfZ-4-k>
- 20) <https://www.youtube.com/watch?v=laG4h127kRo>
- 21) <https://www.youtube.com/watch?v=sgXM33cndnQ>



References

- 22) <https://www.youtube.com/watch?v=cCcuXuDxrJ4>
- 23) https://www.indiamart.com/proddetail/exhaust-valve-spindles-valve-seat-3699308533.html?srsltid=AfmBOorAnKWgaLfGrdl9HjE68Lg1i8qYpv4G_giebq8ixesjYtXG7-qC
- 24) <https://gomechanic.in/blog/what-is-engine-hydrolocking-explained/>
- 25) <https://www.youtube.com/watch?v=OeTjndkQ6LM>